What Survives Diffusion Noise? Exact Scores, Order-by-Order Information Decay, and Transition Learning in Non-Gaussian Markov Chains

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Abstract

Diffusion models are trained to approximate the score of a noised data distribution, a target that is unknown for almost every distribution of interest. We study a family in which it is known exactly and non-trivial: a first-order Markov chain with continuous, non-Gaussian transitions under coordinatewise Ornstein–Uhlenbeck noise, where conditioning leaves the posterior factor graph a tree and one forward–backward sweep of functional message passing returns the exact score. The laboratory separates what diffusion noise does to different orders of structure. The channel attenuates the Hermite moment of degree m by e^{-mt} exactly, so that, for a law with an exponential moment, the Fisher information one noised sequence carries about a feature first visible at degree m decays as e^{-2mt} with an explicit constant: correlation (degree two) as e^{-4t}, non-Gaussian shape (degree four) as e^{-8t}; and the gap between the true score and that of the covariance-matched Gaussian—which we show is exactly what Gaussian message closure computes—is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, with the crossover set by the covariance spectrum. Tree-exact inference verifies the rates and the constants, adding the fourth-order term back to the Gaussian score closes the gap wherever the expansion converges, and heavy-tailed driving noise reaches the same constant. We then learn the transition law from noised sequences alone by expectation–maximisation around the same recursion: the fitted shape sits at the information reference while the coordinate is identifiable and collapses to Gaussian driving noise where the reference says it must, and moment estimators show the predicted rates at a fraction of the efficiency. Against six sequence denoisers trained under one recipe the structured estimator is ahead at every sample size; at the estimator's own information interface—one noised view per sequence, no clean chain, no true score for model selection—the best network is a factor 7.9 behind. The advantage is earned by the non-Gaussian capacity of the driving-noise model, transfers to generation, extrapolates in length, and ends at a misspecification strength that a closed-form bias predicts.

1. Introduction

A diffusion model generates by reversing a noising process, and the reversal is driven by the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ of the noised data law (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021). Training regresses a network onto that score through denoising (Vincent, 2011), so every question about what the network has learned is a question about the distance between what it computes and the score it should have computed. For real data that target is unknown: a large error cannot be told from a hard target, and the solvable alternatives are Gaussian, whose scores are linear, or mixtures, whose nonlinearity lives in a discrete label rather than in any dependence between coordinates.

A laboratory with an exact, non-trivial score. We study a family that is neither. A first-order Markov chain with continuous, non-Gaussian transitions has a globally dependent, nonlinear score, yet under coordinatewise Ornstein–Uhlenbeck noise its posterior factor graph is a tree, so two sweeps of functional message passing return the score exactly (Section 2). The chain's covariance is fixed by two numbers and everything beyond it by the driving-noise law, so Gaussian, Laplace, Student-t, uniform and bimodal driv-

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ing noise, and a one-parameter family interpolating between them, share one covariance exactly and differ only in higher-order structure. The laboratory therefore separates, by construction, what diffusion noise does to correlation from what it does to shape.

Three questions. What information about structured data survives diffusion noise? Can the dynamics that generate the data be learned from noised sequences alone, and how efficiently? And how much does knowing the structure reduce the data needed to recover the score, compared with a network that must discover it?

Three answers. The first is a theorem (Section 3): diffusion noise is a low-pass filter on statistical order. The channel attenuates the Hermite moment of degree m exactly by e^{-mt}; for a law with an exponential moment, the Fisher information a noised sequence carries about a feature first visible at degree m decays as e^{-2mt}, with a constant we give in general, and the number of sequences needed for a fixed local precision grows as e^{2mt}. Correlation lives at degree two and non-Gaussian shape at degree four, so the shape is lost twice as fast. The gap between the true score and that of the covariance-matched Gaussian, which we prove is exactly what single-Gaussian message closure computes (Proposition 1), is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ to leading order, with a crossover $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ that is exact for the Gaussian chain. The exact score lets us verify the exponents and the constants, show that the missing fourth-order term is the whole of the gap above the crossover, and explain, rather than contradict, earlier measurements on this model that stopped before t_*.

The second is empirical (Section 4). We learn the transition law from noised sequences alone by expectation–maximisation around the same exact recursion, with Fisher's identity supplying the exact likelihood gradient from one forward–backward pass. The information curve of Section 3 predicts the level at which the shape of the driving-noise law stops being identifiable from a given number of sequences, and the fits do what it says: they sit at the information reference below that level and collapse to Gaussian driving noise at it, while the correlation survives to the crossover.

The third is a controlled comparison (Section 5). Against six sequence denoisers trained under one recipe the structured estimator is ahead at every sample size; at the estimator's own information interface, one noised view per sequence with neither a clean chain nor the true score for model selection, the best network is a factor 7.9 behind, a larger gap than at the standard interface. The advantage is earned by the non-Gaussian capacity of the driving-noise model and not by the Markov factorisation alone, transfers to generation, extrapolates in sequence length, and ends at a misspecification strength that a closed-form bias predicts. Given enough clean data the networks do pass the second-order floor of Proposition 1, so the gap is a statement about information, not about a barrier they cannot cross.

Contributions. (i) An exactly solvable non-Gaussian, correlated diffusion laboratory with a validated score reference, and the identification of Gaussian message closure with the LMMSE of the covariance twin (Proposition 1), used as an apparatus check at every regime. (ii) Theorems 2, 4 and 5: exact Hermite attenuation and the order-by-order decay of score gaps and Fisher information, with constants, the general order, the exponential-moment hypothesis and what lies beyond it, verified with the exact score. (iii) Transition learning from noised data by exact-inference EM, placed against the information curve level by level, with the collapse of the shape coordinate located and a moment estimator as the rate-correct, inefficient competitor. (iv) A comparison against learned denoisers at three declared information interfaces, from capacity attribution to misspecification with its closed-form bias, showing what explicit structure buys and where it stops.

2. Setup: an exactly solvable non-Gaussian laboratory

Data and channel. A clean sequence $\mathbf{a} \in \mathbb{R}^L$ is a first-order Markov chain,

$$P_0(\mathbf{a}) = \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1}), \quad K(a' \mid a) = p_\varepsilon(a' - \rho a), \tag{1}$$

with $a_1 \sim \mathcal{N}(0,1)$, $|\rho| < 1$, and a driving-noise density p_ε of mean zero and variance $q = 1-\rho^2$. The variance choice makes $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly for every driving-noise law, so the families we use—Gaussian, Laplace, Student-t, uniform, a symmetric two-component mixture, and the generalised Gaussian $p_\varepsilon(\varepsilon) \propto \exp(-|\varepsilon/s_\beta|^\beta)$ whose exponent β moves the excess kurtosis from $+3$ (Laplace) through 0 (Gaussian) to -1 at $\beta = 6$, at fixed variance—share one covariance $\mathbf{\Sigma}_0$ and, every driving-noise law being symmetric, differ only through cumulants of order four and higher. The forward process is the coordinatewise Ornstein–Uhlenbeck channel

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}), \tag{2}$$

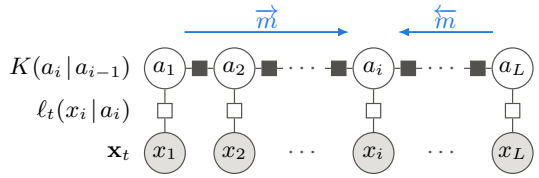


Figure 1. Why the score is exact. Latent sites (white) carry the chain prior through its transition factors (filled squares). Coordinatewise noising attaches one unary factor per site (open squares), joining a_i to its own observation x_i (shaded), and adds no edge among the latent variables. The posterior factor graph is therefore the prior's chain, a tree, so the forward and backward sweeps $\overrightarrow{m}, \overleftarrow{m}$ are exact; meeting at site i they give $\langle a_i \rangle_{x,t}$, and Tweedie's identity returns $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$ exactly. The interior of the chain is elided; L is its length.

so $\mathrm{Var}(x_{t,i}) = 1$ at every t and the noised law P_t tends to $\mathcal{N}(0, \mathbf{I})$.

The score is a posterior mean, and the posterior is a tree. Tweedie's identity gives $S(\mathbf{x}, t) = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$, with $\langle \mathbf{a} \rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$ (Robbins, 1956; Efron, 2011). Conditioning Equation (1) on $\mathbf{x}$ multiplies the prior by one unary factor $\ell_t(x_i \mid a_i) \propto \exp[-(x_i - e^{-t}a_i)^2/2\Delta_t]$ per site: no edge is added among the latent variables, the posterior factor graph is the prior's chain, and sum-product on it is exact (Pearl, 1988; Kschischang et al., 2001). Two sweeps of the functional messages $\overrightarrow{m}_i(a) \propto \int K(a \mid a')\ell_t(x_{i-1} \mid a')\overrightarrow{m}_{i-1}(a')\,\mathrm{d}a'$ and its mirror image $\overleftarrow{m}_i$ (Appendix B, Equation (7)) return the site marginals $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$, and hence the exact score of a correlated, non-Gaussian, L-dimensional law. Messages are functions on $\mathbb{R}$; we carry them on a grid of N_{g} points on $[-A, A]$ with trapezoidal quadrature, exact for the discretised factor model and controlled for the continuous one (at $N_{\mathrm{g}} = 401$, $A = 8$ and the frozen correlation the reference is accurate to 10^{-5} relative on ordinary draws for every family used; the domain, not the spacing, is what a heavy-tailed family tests, and at $\rho = 0$ the Student-t family needs a wider one, Appendix B). Throughout, "the score" means this tree-exact inference, evaluated by validated quadrature.

The covariance twin and the price of second-order closure. The Gaussian chain with the same ρ and q has the closed-form score $S^{\mathrm{G}}(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$, $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$, and posterior mean $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$, the linear minimum-mean-square (LMMSE) estimator.

Proposition 1 (Gaussian closure computes the LMMSE of the covariance twin). Let Π_{G} project a density onto the Gaussian with its mean and variance, and let moment-projected sum-product apply Π_{G} to each message after every transition integral in Equation (7). For the chain Equation (1) with $\rho \neq 0$ and any driving-noise law of mean zero and variance q, it returns exactly the LMMSE estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$, the same for every such law.

The proof (Appendix A) is an induction over the two sweeps: a convolution's first two moments depend on the driving noise only through its first two moments, so every projected message equals the Gaussian chain's message. Two consequences are used as checks below. Sum-product with any Gaussian kernel is a linear map of $\mathbf{x}$, and the LMMSE of the true covariance minimises the posterior-mean error over linear maps, so the LMMSE floor bounds every linear estimator; and the Gaussian quasi-likelihood of noised data depends on the data only through their sample covariance, whose expectation is $e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I}$, so a linear-Gaussian kernel fitted by maximum likelihood is consistent for (ρ, q) and lands on that floor. Proposition 1 thus turns the gap between S and S^{G} into an interpretable object: it is precisely what any estimator that carries only second-order structure—a linear-Gaussian kernel, a linear denoiser, or the Kalman smoother (Proposition 7)—must leave on the table. Section 3 computes its size.

3. What survives diffusion noise: attenuation by statistical order

Write $\mathcal{N}$ for the standard Gaussian on $\mathbb{R}^L$, He_k for the probabilists' Hermite polynomials and $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$ for their products, of degree $|\nu|$, with $\mathbb{E}_{\mathcal{N}}[H_\nu H_\mu] = \nu!\,\delta_{\nu\mu}$. The channel Equation (2) acts on Hermite moments diagonally.

Theorem 2 (Hermite attenuation). For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$, $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$.

Proof. Mehler's identity $\mathbb{E}_z[\mathrm{He}_n(\alpha a + \sqrt{1-\alpha^2}\,z)] = \alpha^n \mathrm{He}_n(a)$ for $z \sim \mathcal{N}(0,1)$ follows from the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x)s^n/n!$; coordinates are independent given $\mathbf{a}$, and $\alpha = e^{-t}$. □

Nothing here requires $P_0/\mathcal{N} \in L^2(\mathcal{N})$—which fails for the Laplace chain and, because $\lambda_{\max}(\mathbf{\Sigma}_0) > 2$, for the Gaussian chain at $\rho = 0.85$—only finite moments. Two consequences are exact at every t: cumulants of order $m \geq 3$ scale as e^{-mt} (Gaussian noise has none), and the degree-4 Hermite moments of two variance-matched laws differ exactly by their fourth-cumulant tensors κ_4, attenuated by e^{-4t}. Three clocks. For a feature first visible at Hermite degree m, its amplitude decays as e^{-mt} (exact, Theorem 2); the Fisher infor-

mation a sequence carries about it, a squared amplitude, as e^{-2mt} (Theorems 4 and 5); and the number of sequences needed for a fixed local precision grows as e^{2mt} (Corollary 6). Correlation lives at degree two: e^{-2t}, e^{-4t}, e^{4t}. Non-Gaussian shape first appears at degree four: e^{-4t}, e^{-8t}, e^{8t}. Diffusion noise is a low-pass filter on statistical order.

The consequence for scores and for learning is an expansion in the signal-to-noise ratio e^{-2t}. Let $s = e^{-t}\mathbf{x}/\Delta_t$, $b = e^{-2t}/\Delta_t$ and $\Lambda(s,b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$, so that $\log P_t(\mathbf{x}) = \log \mathcal{N}_{\Delta_t}(\mathbf{x}) + \Lambda(s,b)$ and $S = -\mathbf{x}/\Delta_t + (e^{-t}/\Delta_t)\nabla_s \Lambda$ exactly. Grading Λ by order in e^{-t}, its second-order part is the same for P_0 and its covariance twin, and its fourth-order part differs by a term linear in κ_4. The expansion is legitimate when Λ is finite in a neighbourhood of the origin, which is the one hypothesis the theorem needs:

Assumption 3 (Exponential moment). $\mathbb{E}_{P_0} e^{\lambda|\mathbf{a}|} < \infty$ for some $\lambda > 0$.

Every driving-noise law used below satisfies it except Student-t, for which $\Lambda(s,0) = +\infty$ at every $s \neq 0$; that family is discussed after Theorem 5.

Theorem 4 (Order-by-order information decay). Let P_0 satisfy Assumption 3 with covariance $\mathbf{\Sigma}_0$, let S^{G} be the score of its covariance twin, and fix L. As $t \to \infty$,

$$\mathbb{E}_{P_t}\|S - S^{\mathrm{G}}\|^2 = \tfrac{1}{6}\|\kappa_4\|_F^2\, e^{-8t}(1+\epsilon_t), \tag{3}$$

$$I_t(\theta) = \tfrac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2\, e^{-4t}(1+\epsilon_t) \text{ if } \partial_\theta \mathbf{\Sigma}_0 \neq 0, \tag{4}$$

$$I_t(\theta) = \tfrac{1}{24}\|\partial_\theta \kappa_4\|_F^2\, e^{-8t}(1+\epsilon_t) \text{ if } \partial_\theta \mathbf{\Sigma}_0 = 0, \tag{5}$$

the last with $\partial_\theta \kappa_4 \neq 0$, and with $\epsilon_t = O(e^{-2t})$, where $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ is the Fisher information one noised sequence carries about a parameter θ of P_0, and $\|\cdot\|_F$ is the Frobenius norm of the matrix or symmetric 4-tensor over ordered indices. For a Gaussian chain Equation (4) is exact at every t with $I_t = \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_t^{-1}\partial_\theta \mathbf{\Sigma}_t \mathbf{\Sigma}_t^{-1}\partial_\theta \mathbf{\Sigma}_t)$ and $\epsilon_t = O(e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0))$ explicitly.

The information statements Equations (4) and (5) are the diffusion-time form of a principle established for low-SNR latent-variable models: the log-likelihood expands in powers of the SNR with the order-k term carrying k-th-moment information (Katsevich & Bandeira, 2023; Perry et al., 2019; Bandeira et al., 2023), and Fisher information about directions first identifiable at moment order k scales as SNR^k (Balanov et al., 2026); with $\mathrm{SNR}_t \simeq e^{-2t}$ these are Equations (4) and (5). The cumulant attenuation behind Theorem 2 is also written down for the forward process by Aarts et al. (2025). What this paper adds is the score-level statement Equation (3) with its constant, made interpretable by Proposition 1; the general order and its constant below; the hypothesis under which the expansion is a theorem and what happens outside it; and the exact verification and learning consequences that follow. The proof (Appendix A) expands Λ to fourth order on the bulk $\{|s| \le \lambda/2\}$, where Assumption 3 makes it analytic with uniformly bounded tilted moments, takes the $\mathcal{N}$-second moment of the leading Hermite term by Wick's theorem, and bounds the tail, whose probability is $\exp(-c\,e^{2t})$ under Assumption 3; the scalar leading term of Equation (3) is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$. The leading orders of a degree-2 and a degree-4 coordinate do not couple, because He_2 and He_4 are $\mathcal{N}$-orthogonal, so the efficient information of a shape coordinate after lower-degree nuisance coordinates are estimated has the same rate and constant as its raw information; a nuisance of the same degree costs an $O(1)$ factor.

Theorem 5 (General order). Let P_0^θ satisfy Assumption 3 uniformly near θ_0, and let m be the smallest degree at which $\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})] \neq 0$ for some $|\nu| = m$. Then, at fixed L,

$$I_t(\theta) = e^{-2mt} \sum_{|\nu|=m} \frac{\big(\partial_\theta \mathbb{E}_{P_0}[H_\nu(\mathbf{a})]\big)^2}{\nu!}\big(1 + O(e^{-2t})\big).$$

The constant is the Hermite norm of the derivative of the degree-m moment tensor; for $m = 2$ it is $\frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$ and for $m = 4$ at fixed covariance $\frac{1}{24}\|\partial_\theta \kappa_4\|_F^2$, since the multiset ν collects $4!/\nu!$ ordered index tuples. A direction with no non-vanishing Hermite footprint is not identifiable at any noise level; under Assumption 3 the law is determined by its moments, so such a direction is null.

Corollary 6 (Sample complexity by order). Fix t and let θ be a coordinate of lowest degree m with efficient information $I_t^{\mathrm{eff}}(\theta)$. For M i.i.d. sequences observed at level t and any estimator sequence, the local asymptotic minimax risk satisfies $\lim_{c\to\infty} \liminf_M \sup_{|\theta'-\theta| \le c/\sqrt{M}} M\, \mathbb{E}_{\theta'}(\hat{\theta} - \theta')^2 \ge 1/I_t^{\mathrm{eff}}(\theta) = \Theta(e^{2mt})$, and maximum likelihood attains it. A covariance coordinate therefore needs $M \propto e^{4t}$ sequences for a fixed local precision, a shape coordinate $M \propto e^{8t}$; the ratio of the two informations is $\propto e^{-4t}$.

This is Hájek–Le Cam for a regular parametric model at fixed t; it says nothing uniform in t (the limits do not commute), nothing about absolute precision for a bounded coordinate once $1/\sqrt{MI_t}$ exceeds its range, and nothing about an optimiser at a finite budget. All three caveats are visible in Section 4.

Where the asymptote begins. For the Gaussian chain the expansion parameter is $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0)$, with $\lambda_{\max} \le (1+\rho)/(1-\rho)$, and $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ is its scale, exact for the Gaussian chain in the sense that $P_t/\mathcal{N} \in L^2(\mathcal{N})$ iff $e^{2t} > \lambda_{\max} - 1$, that is for $t > \frac{1}{2}\log(\lambda_{\max}-1)$, just below t_*; below t_* the long-wavelength covariance modes are still of order one and the rates are shallower. For non-Gaussian laws the same scale is a heuristic that the measurements below support. At $\rho = 0.85$, $t_* = 1.17$: a strongly correlated chain keeps its correlation information longer, and any measurement stopped near t_* sees slopes short of -4 and -8.

Heavy tails. Student-t_ν driving noise violates Assumption 3. For a regularly varying tail of index ν the expected squared gap acquires a term of order $e^{-\nu t}$ carried by sequences with $|\mathbf{a}| \gtrsim e^t$, whose probability is itself of order $e^{-\nu t}$: in expectation the tail term leads once it exceeds $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, and on any sample of practical size the Hermite law is all that is seen, because the events that carry the tail term are never drawn. Both halves are visible in the numbers (Appendix K): for the scalar t_5 law an untruncated quadrature gives a local slope of -5.28 on the last interval and a ratio to the Hermite term of 9416 at $t = 5$, while the t_5 chain sampled with 8192 sequences on a domain wide enough to hold them sits at 1.03 $[0.99, 1.09]$ of the constant at $t = 4$ (Figure 2c). We state the heavy-tailed rate as an observation supported by quadrature, not as a theorem; the moment condition of Theorem 4 is not merely technical.

Verification with the exact score. Figure 2 tests Theorem 4 where nothing has to be approximated. For the Gaussian chain the closed form gives slope -3.97 above t_* at $\rho = 0.85$; for the non-Gaussian chain the shape information, estimated from per-sequence scores through the tree-exact recursion, falls with slope -7.94 ($\beta = 2$) and -8.01 ($\beta = 3$) above t_* against the predicted -8, and the correlation information with slope -3.98 against -4. The constants are verified at two levels of precision: in the scalar case an untruncated quadrature reproduces the shape constant of Equation (5) to within 0.6% and the covariance constant of Equation (4) to five digits (Appendix K), while on the chain the outer-product estimate of the shape constant reaches it only to within twenty per cent before its quadrature floor; the closure gap of the Laplace chain decays with slope -7.93 and reproduces $\frac{1}{6}\|\kappa_4\|_F^2$ to 0.99 $[0.91, 1.08]$ at $t = 4$ from 8192 sequences (the per-sequence gap is heavy-tailed, which is why so many are needed). The theorem also explains a measurement the previous literature on this model reported as a discrepancy: slopes of about -3 and -6 obtained over $t \le 1.6$ are the pre-asymptotic values below $t_* = 1.17$, not a failure of the law. Finally, Equation (3) says what the covariance twin lacks: adding the degree-4 Hermite term back, the first Edgeworth correction with the true $\mathbf{\Sigma}_0$ and κ_4, closes 0.961 of the squared gap at $t = 0.68$ and all of it from $t = 2$ on every family with an exponential moment, and diverges below $t = 0.15$ (Appendix K); above the crossover a score built from the first four moments is the score.

What this says about score learning. An estimator that carries only second-order structure cannot go below the LMMSE floor $\|S - S^{\mathrm{G}}\|$, which on the Laplace chain at $\rho = 0.85$ is 0.168 of the score energy pooled over the twelve-level schedule and 0.28 at the lowest level (Appendix D). A learner that recovers the fourth-order structure has, by Corollary 6, e^{4t} fewer effective samples per sequence at level t than one that recovers only covariance: the twelve-level schedule is, for shape, effectively a four-level one. Both facts organise the experiments of Section 5.

4. Learning the dynamics through noise

Now drop the assumption that K is known. We observe M noised sequences $\mathbf{x}^\mu$, each at one level t_μ, never the clean chain, and maximise the marginal log-likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_t[t_\mu, \theta](\mathbf{x}^\mu)$ of a parameterised kernel K_θ.

The exact gradient from one sweep. Because only the transition factors depend on θ, Fisher's identity (Cappé et al., 2005) gives, on the grid,

$$\nabla_\theta \mathcal{L}(\theta) = \big\langle \mathbf{\Xi}(\theta),\, \nabla_\theta \log K_\theta \big\rangle, \qquad \mathbf{\Xi}_{kl} = \sum_\mu \sum_{i \ge 2} P(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu), \tag{6}$$

where the expected transition mass $\mathbf{\Xi}$ is read off the pairwise marginals that the same forward–backward pass returns. No differentiation through the recursion is needed, and the identity is verified against finite differences and against brute-force enumeration of the evidence on short chains (tests/test_em_bp.py, tests/test_fisher_bruteforce.py). $\mathbf{\Xi}$ is also all the M-step of EM needs: for a Gaussian kernel the maximiser is the belief-weighted Yule–Walker pair $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$ with $S_{00} = \sum_{kl}\mathbf{\Xi}_{kl}u_l^2$, $S_{01} = \sum_{kl}\mathbf{\Xi}_{kl}u_k u_l$, $S_{11} = \sum_{kl}\mathbf{\Xi}_{kl}u_k^2$ and $W = \sum_{kl}\mathbf{\Xi}_{kl}$ (k indexes the child site, l the parent); for the mixture kernel $K_\theta(a' \mid a) = \sum_{c=1}^{C}\pi_c\,\mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)$ used below, the complete-data problem carries a second latent label and we take conditional-maximisation sweeps, a

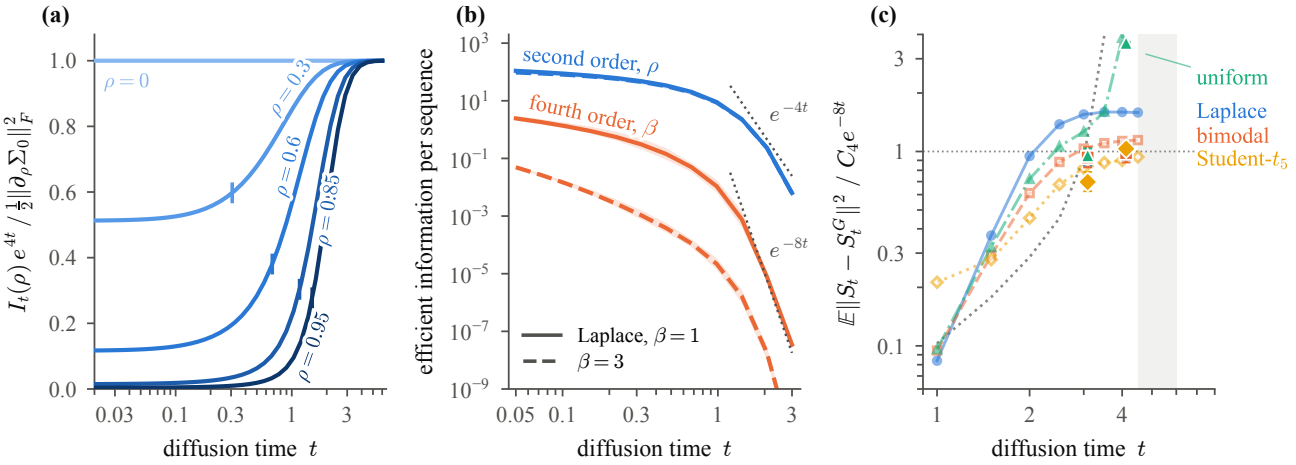


Figure 2. What survives diffusion noise. (a) Exact Fisher information of the Gaussian chain about ρ, scaled by its predicted asymptote $\frac{1}{2}\|\partial_\rho\mathbf{\Sigma}_0\|_F^2 e^{-4t}$, for five ρ; every curve reaches 1, and the tick on each marks $t_*(\rho)$. (b) Efficient information per noised sequence about the second-order mode, the correlation ρ, and the fourth-order mode, the shape β, of the generalised-Gaussian chain, from tree-exact inference and Fisher's identity, 512 sequences $\times$ 8 replicates (bands: replicate range); guides e^{-4t} and e^{-8t}. Above t_* the fitted slopes are -4.00 (ρ) and -8.01 (β, at $\beta=3$). (c) The price of second-order closure: $\mathbb{E}\|S-S^{\mathrm{G}}\|^2$ measured by tree-exact BP, divided by the predicted $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, for four driving-noise families on a domain wide enough for their samples (Appendix B). Thin curves: 512 sequences, paired across levels; bold markers with bars: the constant from 8192 sequences with a bootstrap interval at $t=3$ and 4; grey dotted: the same Student-t_5 sequences on the operating $(801, 8)$ domain, where the truncated prior drives the reference. The shaded band is where the double-precision floor of the posterior means enters; the uniform family, with the smallest constant, reaches it first. Laplace, the bimodal mixture and Student-t_5 reach the constant.

generalised EM whose ascent is monotone (Dempster et al., 1977; Meng & Rubin, 1993). The estimator sees one frozen noised view per sequence and never a clean trajectory; each reported fit selects its iteration count on a validation bundle disjoint from the test bundle.

The information curve predicts what is learned at each level. Corollary 6 makes a prediction about learning that the exact score does not need: fitting from M sequences all observed at level t, the shape of the driving-noise law stops being identifiable at the level where its Cramér–Rao reference crosses the coordinate's range, while the correlation survives to the crossover. Figure 3 measures both with $M = 512$ Laplace-chain sequences at $\rho = 0.85$ and 16 replicates per level, against the information reference of Section 3: the Cramér–Rao bound $1/\sqrt{M\,I_t}$ for the correlation and, by the chain rule at the Laplace member of the shape family, for the excess kurtosis. The reference is for an unbiased estimator in a one-parameter family and the estimator is an eight-component mixture with its iteration count selected on validation, so the comparison is a yardstick, read level by level. Up to $t = 0.4$ every fit recovers the shape and the dispersion of $\widehat{\kappa_4}$ across replicates is within the sampling error of the reference (0.69 against 0.838 at $t = 0.4$; every level in Appendix K). At $t = 0.7$ the fits split, 4 of 16 collapsing to Gaussian driving noise and the rest shrunk; at $t = 1$, 15 of 16 collapse, and the reference has crossed the full range of the coordinate at $t = 0.98$: the level at which the information runs out is the level at which the estimator stops estimating. The correlation is recovered to 0.0092 at $t = 1$ against a reference of 0.0151 and is lost only past the crossover $t_* = 1.17$, 0.0325 at $t = 1.4$ and 0.3430 at $t = 2$, of which part is budget rather than information: with the em ladder extended to 8000 iterations on the same seeds and data (e06c; hollow markers) the two fall to 0.0215 and 0.2441 while nothing below $t = 1$ moves by more than 0.0009; past the crossover em's contraction slows with the information it has. The same coordinates estimated by the method of moments—the lag-one product for ρ and the per-site fourth cumulant divided by e^{-4t} for κ_4, whose marginal statistics are unbiased by Theorem 2 up to the $O(e^{4t}/M)$ plug-in bias of a sample cumulant—are 5–6 times worse for ρ and 2.4–5.5 times worse for κ_4 at the informative levels, never collapse, and show the asymptotic growth where the estimate is already useless (log-slopes 1.83 and 3.73 on $t \geq 1$ against 2 and 4; Appendix K): exact inference buys the efficiency below the crossover, and at high noise the two estimators share the rate, since the likelihood score of a degree-m coordinate is itself a degree-m Hermite statistic, of which the lag-one and per-site statistics used here keep only a part. Below the crossover both errors grow more slowly than the asymptotic rates (fitted log-slopes 1.29

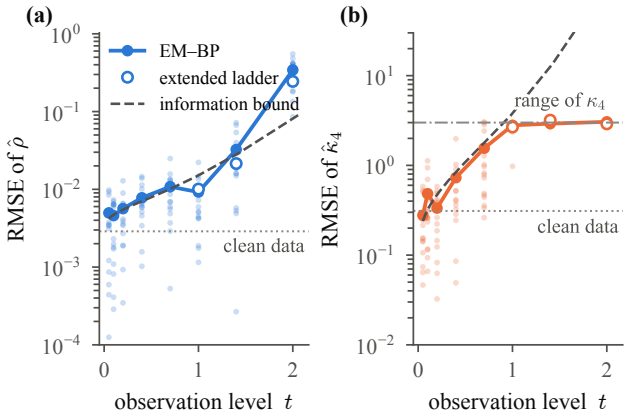


Figure 3. Learning through noise. Recovery of the correlation coordinate and of the driving-noise shape from $M = 512$ sequences observed at a single level t, sixteen replicates (points: individual fits; line: RMSE), against the Cramér–Rao reference (dashed), the clean-data floor (dotted) and, for the shape, the coordinate's range. The fits are unimodal and at the reference at the lowest levels, split into a shrunk cluster and Gaussian-collapsed fits from $t = 0.7$, and collapse where the reference crosses the range. Hollow markers: the same fits with the em ladder extended to 8000 iterations (e06c).

and 3.03 on $[0.2, 0.7]$ against 2 and 4), as Section 3 predicts; the asymptotic window lies beyond the reach of $M = 512$ sequences at this correlation, because the shape collapses before t_*. The clean-data fit gives the optimiser's own floor, 0.0029 and 0.310. This is also the arithmetic behind the one-view protocol: of twelve levels spread over $t \in [0.05, 3]$, only the lowest four carry appreciable information about shape.

5. What explicit structure buys, and where it stops

This section answers the third question in three steps: what explicit structure buys and why (capacity), whether the advantage is the structure or the data the estimator is given (three information interfaces, and a comparison at EM–BP's own), and where it stops (length, generation, misspecification). All comparisons use the frozen configuration of Appendix C: Laplace chain, $\rho = 0.85$, $L = 32$, twelve log-spaced levels on $[0.05, 3]$, and the schedule-pooled relative score error against the tree-exact reference. Paired seeds are the inferential unit, sixteen at the frozen configuration and eight in the sweeps across regimes; every selection is made on development seeds or on a validation bundle that never enters a reported number. Three information interfaces are distinguished throughout: (A) the standard denoising recipe, clean sequences with a fresh corruption at every step; (B) clean sequences with one fixed corruption each; (C) one noised view per sequence and nothing else, which is what EM–BP sees. Three reference points bracket every estimator (Appendix D): the isotropic score $-\mathbf{x}$ at 0.670, the LMMSE oracle with the true covariance at 0.168, the floor of every second-order estimator by Proposition 1, and the exact score at zero.

Table 1. Schedule-pooled relative score error on all sites, mean over 16 seeds (s.e. in the text macros; the isotropic reference is 0.669 at every size). Every arm is scored on identical test bundles. The last column is the same capacity fitted to the driving-noise law itself—the C-component mixture closest in Kullback–Leibler divergence, an infinite-data reference for the family rather than a bound; † marks fits whose narrowest component is under two grid cells.

	sequences M				
kernel	32	128	512	2048	$M = \infty$
LMMSE floor	0.167	0.167	0.167	0.167	0.167
$C = 1$	0.174	0.168	0.168	0.167	0.167
$C = 2$	0.084	0.050	0.035	0.028	0.039
$C = 4$	0.093	0.053	0.037	0.025	0.0057
$C = 8$	0.099	0.055	0.036	0.023	0.0022†
$C = 16$	0.100	0.056	0.036	0.022	0.0007†

Capacity: the margin is fourth-order. Table 1 places every arm on identical test bundles. A linear-Gaussian kernel ($C = 1$) lands on the LMMSE floor, within sampling error, at every size (0.174, 0.168, 0.168, 0.167 against the floor 0.167–0.167), as Proposition 1 requires; the mixture kernel goes far below it, reaching 0.023 at $M = 2048$, 7.9× under the floor. The paired $C = 8$ versus $C = 1$ difference is 0.144 (bootstrap $[0.141, 0.147]$) at $M = 2048$ and 0.075 ($[0.059, 0.090]$) already at $M = 32$, in favour of the mixture in 16 and 16 of 16 seeds. At $M = 2048$ the $C = 8$ arm still selects the top of its 1200-iteration ladder in 10 of 16 seeds, so its error there is an upper bound on what a longer ladder would give. The fitted driving noise excess kurtosis is 2.89 at $M = 2048$ against the true 3; 1 of 320 fitted cells fail the two-grid-cell resolution gate and are reported, not dropped. The last row of Table 1 is the family's infinite-data reference, the C-component mixture closest in Kullback–Leibler divergence to the Laplace driving noise: its $C = 1$ member scores 0.167, the LMMSE floor to three decimals, and at $C = 8$ it scores 0.0022, so the fitted kernel at $M = 2048$ sits 10.3× above what its own family can express, a data limit and not a capacity limit (Appendix F). A linear-Gaussian kernel ($C = 1$) lands on the LMMSE floor, as Proposition 1 requires; the mixture kernel goes far below it. An earlier study of this model concluded the opposite from a sweep that, it turned out, had fitted every arm to a single training chain (Appendix F); with

independent chains and the checkpoint selected identically for every capacity the effect is unambiguous. Capacity buys exactly what the closure discards: across the generalised-Gaussian family at fixed covariance the paired gain of $C = 8$ over $C = 1$ is ordered by κ_4, 0.131 at the Laplace member and a small loss, -0.008, at the Gaussian member, where there is no fourth-order structure to buy (Appendix G, Figure 7). The contrast is not a property of the frozen point: in all 15 cells of a sweep over $\rho \in [0.3, 0.95]$ and $L \in [16, 64]$ the mixture beats the Gaussian kernel with the paired interval excluding zero, the paired difference $C = 8$ minus $C = 1$ running from -0.253 at the most favourable corner to -0.047 at the least, and in every cell the $C = 1$ arm lands on the LMMSE floor to 0.0099, which checks the whole apparatus away from the frozen configuration (Appendix H).

Baselines under one recipe. Table 2 compares six architectures trained under one recipe (interface A) with the configuration that won each family's screen on the development seeds (Appendix C). Weight sharing separates the fully connected network from the rest at every size (0.517 against 0.186 at $M = 32$); locality is the better prior when sequences are scarce, the window head being the best network at $M = 32$, and stops paying once they are not: the window head moves only from 0.121 to 0.118 between $M = 512$ and 2048 while the global architectures keep improving, the best reaching 0.060. That plateau is the architecture's and not the radius's: the best estimator of any kind that sees only a window of radius r is $\mathbb{E}[a_i \mid x_{i-r}, \ldots, x_{i+r}]$, computable by sum-product on the sub-chain, and at radius 4 its pooled error is 0.030 against the head's 0.118 (Appendix K, Figure 8). Given enough clean sequences the networks pass the second-order floor, 5 of six sitting below 0.167 at $M = 2048$ (Proposition 1); nothing intrinsic holds them at second order. The structured estimator is ahead at every size nonetheless, 0.023 against 0.060 at $M = 2048$, a paired factor 2.83 ([2.37, 3.42], 16 of 16 seeds). That factor is an upper bound on the advantage at this interface, since the largest networks were still improving when their ladder ended (88% of the transformer cells at the last rung), and it is not information-matched: the networks see clean sequences and a fresh corruption at every step, EM–BP one noised view per sequence and no clean chain at all.

The comparison at matched information. Interface C is EM–BP's own: one noised view per sequence split across the twelve levels, no clean chain, and, since the true score is unavailable for selection there, the checkpoint chosen on the implicit score-matching objective of a held-out noisy bundle. Trained by implicit score matching on exactly that bundle, the window head reaches 0.165 at $M = 2048$ against EM–BP's 0.023, a paired factor 7.9 ([6.6, 9.3], 16 of 16 seeds); the transformer reaches 0.421. Under interface B, the clean sequence supplied as a regression target but with one fixed corruption per sequence, the same head reaches 0.155, a factor 7.3: even given strictly more than EM–BP sees, one corruption per sequence is not enough. The factor that remains at equal information is what the factorisation and the exact E-step are worth, and it is larger than the factor at the standard interface, not smaller. EM–BP's own selection needs no true score either: choosing its iteration count by held-out one-view log-evidence gives 0.105 against 0.105 with oracle validation at $M = 32$, where the rung matters most, the two rules agreeing in 7 of 8 seeds (Appendix K).

The matched comparison away from the frozen point. The factor is not a property of the frozen corner. Re-running the matched comparison across $\rho \in \{0.5, 0.85, 0.95\}$ and $L \in \{32, 128\}$, with the networks re-screened inside each regime and 16 seeds per cell (Appendix J), EM–BP with its iteration count chosen by held-out evidence, so that no true score enters either arm, is ahead of the best implicit-score-matching network in all 6 cells and in every seed: by a paired factor 5.7 [4.9, 6.7] at $\rho = 0.5$, $L = 32$, and 8.3 [6.6, 10.3] at the frozen point, consistent with the sixteen-seed 7.9 above. The factor is smallest at $\rho = 0.5$ at both lengths, where the stationary marginal carries the most fourth-order structure (1.80 against 0.15 at $\rho = 0.95$), which is the direction Section 3 gives; beyond $\rho = 0.85$ the two lengths part company, the factor rising further at $L = 32$ and falling back at $L = 128$, so we report the direction below $\rho = 0.85$ and no trend above it. Lengthening the chain does not collapse the advantage (4.6 to 8.0 at $L = 128$). Given the clean sequence as a regression target (interface B) the best network is still a factor 7.6 behind at the frozen point, and the factors are conservative throughout: the evidence-selected EM–BP selects the last rung of its ladder in 56 of 96 cells, so its error there is an upper bound. Against the second-order floor the sweep separates two questions that coincide at the frozen configuration. The best network's error changes little across it (0.089 to 0.196) while the floor falls steeply with ρ, so the network passes the floor where the floor is high ($(0.5, 32), (0.5, 128), (0.85, 128)$) and not where it is low ($(0.85, 32), (0.95, 32), (0.95, 128)$): at $\rho = 0.95$ it is above by $+0.110$ [$+0.108$, $+0.112$] at $L = 32$, against an exact radius-8 floor of 0.025 at that correlation. Where the covariance carries most of the score energy, one noised view per sequence is not enough

Table 2. All-site schedule-pooled relative score error, mean over the confirmatory seeds (± s.e. in the macros), every arm on identical test bundles. Oracles and EM–BP ($C = 8$) from the capacity experiment; networks under the common recipe with the screened configuration.

M	isotropic	LMMSE	MLP	window	TCN	bi-GRU	transf.	U-Net	EM–BP
32	0.669	0.167	0.517	0.186	0.223	0.201	0.231	0.227	0.099
128	0.669	0.167	0.391	0.141	0.139	0.121	0.151	0.150	0.055
512	0.669	0.167	0.276	0.121	0.096	0.078	0.097	0.095	0.036
2048	0.669	0.167	0.208	0.118	0.068	0.060	0.063	0.066	0.023

for a network to recover even the covariance, while the estimator that knows the factorisation recovers it. The same cell with platykurtic driving noise gives the same answer: on the symmetric two-component mixture, whose excess kurtosis is negative at the same covariance (Appendix D), the paired factor is 8.8 [7.4, 10.2] (Appendix J). What the structure buys is fourth-order structure of either sign, not heavy tails.

Length extrapolation. Fitted at $L = 32$ and evaluated against the exact score of longer chains (Table 6), the EM–BP kernel holds 0.023 at $L = 32$ and 0.023 at 256: the law is local, so inference at any length is the same recursion. The convolutional and recurrent networks improve slightly, as longer chains have proportionally fewer boundary sites; the transformer with learned relative positions degrades from 0.063 to 0.703, a failure of the positional encoding beyond its training length rather than of the score it learned (Press et al., 2022; Kazemnejad et al., 2023).

Generation. Samples are drawn by the same reverse SDE (400 steps on a geometric grid, one Brownian path shared by every arm, 16 seeds) from the exact score, the fitted kernel, the LMMSE twin and the isotropic score, and compared with a reference sample of the true law at $t_{\min}$ (Table 5). The exact arm's log-likelihood gap, 0.070 nats per sequence at $M = 2048$, is the sampler's own discretisation error, the reference against which the other arms are read; the fitted kernel sits at 0.164, the LMMSE twin at 1.766 and the isotropic score at 46.295. The excess kurtosis of the generated sequences' autoregressive residuals at $t_{\min}$, where the residual noise lowers even the true law's value below the driving noise's 3, is 1.803 for the fitted kernel against 1.94 for the true law and -0.003 for the twin, which by construction generates Gaussian driving noise: what the second-order estimator discards in score error it discards in the samples.

Misspecification. Every departure is Gaussian so that its score is exact, and every arm is scored against that score; the network arm is a radius-2 window head under the recipe of Appendix C. The ordering degrades smoothly in each mechanism and the strength at which it breaks even is a property of the mechanism, not a single number. A rank-one global latent never reverses the ordering in the swept range: the ratio falls only from 15.55 to 3.97 across the swept range, up to $\delta = 1$. A second-lag coefficient reverses it between 0.3 and 0.4 at both sizes, and a site-dependent ρ between 0.1 and 0.14 at $M = 2048$. Long-range precision coupling does not reverse it anywhere we swept: the ratio decays to 1.00 at $\delta = 0.4$, a tie rather than a crossing. We report brackets and not fitted crossings, because with 16 seeds the paired log-ratio is resolved to a few thousandths and the crossing sits between grid points either way.

The two sizes separate the statistical gain of the structure at small M from its asymptotic bias at large M, and the bias has a closed form: every truth is Gaussian and the fitted driving noise is Gaussian to three decimals, so the $M \to \infty$ limit of EM–BP is the Kullback–Leibler projection of the noised truth onto the linear-Gaussian family, whose relative score error $b(\delta)$ is a trace formula (Appendix K). It reproduces the measured EM–BP error at $M = 2048$ to within 0.019 across every mechanism and strength, $\sqrt{b(\delta)^2 + v/M}$ predicts both sizes (Figure 4, bottom row), and its order in the strength says which mechanisms bite: $b \propto \delta$ for the three mechanisms that perturb of order L directions of the covariance and $b \propto \delta^2$ for the rank-one latent, which perturbs one. Past the break-even the chain estimator has converged to a wrong law and more data buys nothing back, exactly the behaviour Section 3 attributes to a model whose bias, not whose variance, is binding.

6. Related work

Solvable diffusion and the Gaussian part of the score. Diffusion and score matching are due to Sohl-Dickstein et al. (2015); Ho et al. (2020); Song et al. (2021) and Hyvärinen (2005); Vincent (2011); the score–posterior-mean identity is Tweedie's (Robbins, 1956; Efron, 2011). Exactly solvable targets have been Gaussian (Pierret & Galerne, 2024), mean-field or mixture-like

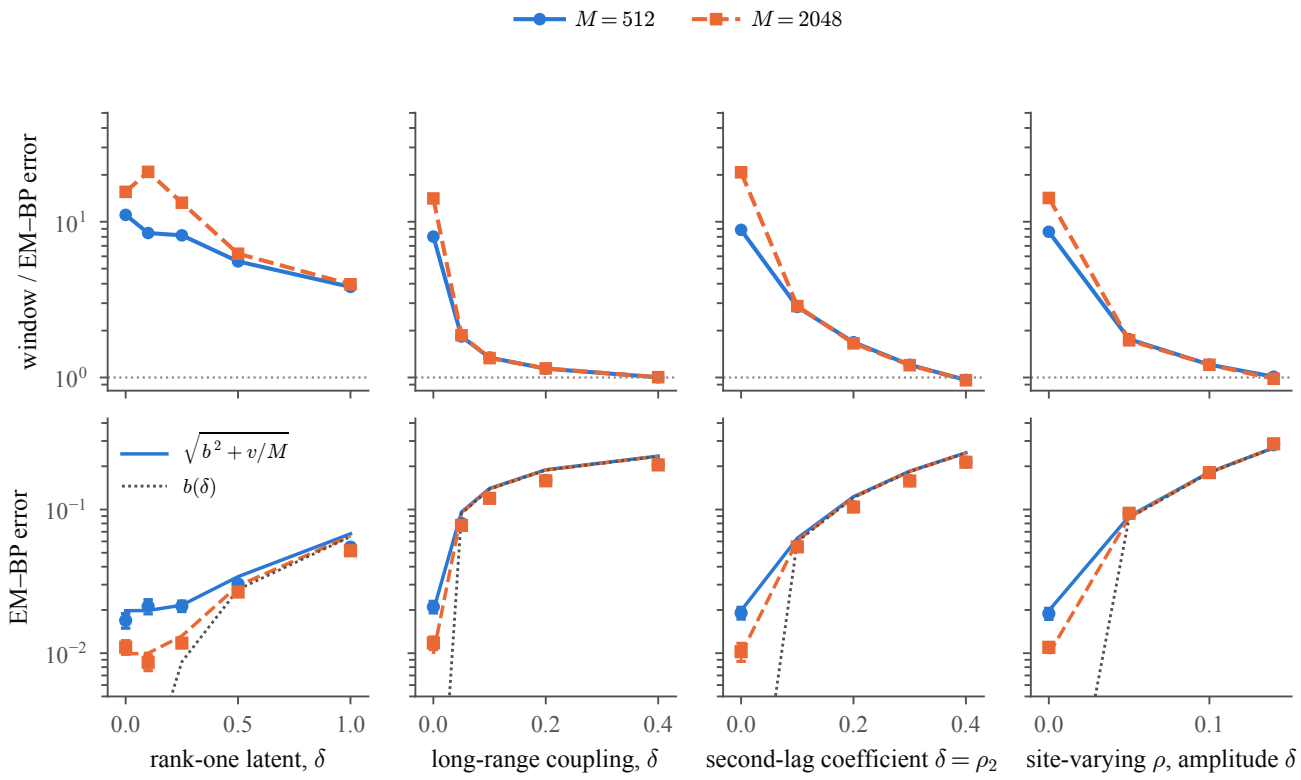


Figure 4. Where the structure stops helping. Ratio of a radius-2 window head's error to EM–BP's against the misspecification strength δ, for four Gaussian departures from the chain with exact reference scores (rank-one global latent, long-range precision coupling, a second-lag coefficient at fixed lag-one correlation, site-dependent ρ), at two sample sizes (top row). Above the dotted line the misspecified chain estimator still wins. Bottom row: EM–BP's own error at both sizes (points, paired standard errors) against the closed-form bias $b(\delta)$ of the Kullback–Leibler projection (dotted) and the finite-sample prediction $\sqrt{b(\delta)^2 + v/M}$ (lines) with v read from the well-specified cells (Appendix K).

(Biroli & Mézard, 2023; Biroli et al., 2024; Cui et al., 2025), or hierarchical with discrete leaves (Sclocchi et al., 2025), on which U-Nets and transformers implement belief propagation (Mei, 2024; Mei & Wu, 2023; Garnier-Brun et al., 2024). A recurrent empirical observation is that learned scores are dominated by their Gaussian part over a wide noise range (Wang & Vastola, 2024; Li et al., 2024; Kadkhodaie et al., 2024; Wang et al., 2026), and that downstream statistics depend on the first two moments alone (Ghane et al., 2025). Our chain is the degenerate tree of depth L with continuous non-Gaussian states; the Gaussian part of its score is the Kalman smoother of the covariance twin (Proposition 1), and Equation (3) is the sharp rate, with its constant and crossover, behind the dominance those papers report.

What survives noise: the low-SNR literature. Theorem 4 belongs to the low-SNR expansions of estimation theory. In Gaussian channels the mutual information and the MMSE expand in the SNR with fourth-order statistics entering at second order, for arbitrary input laws (Prelov & Verdú, 2004; Guo et al., 2005; Dytso et al., 2023; Kong et al., 2023); for latent-variable models the log-likelihood expands with the order-k term matching k-th moments (Katsevich & Bandeira, 2023), which yields the σ^{2k} sample complexity of multireference alignment and orbit recovery (Perry et al., 2019; Bandeira et al., 2023) and, most closely, the layered Fisher-information geometry of Balanov et al. (2026), where information about directions first identifiable at moment order k scales as SNR^k and the method of moments is almost efficient. Under the OU channel $\mathrm{SNR}_t \simeq e^{-2t}$, so that statement and our e^{-2mt} are the same law in different clocks. Aarts et al. (2025) write the forward-process cumulant attenuation e^{-nt} and show that score networks regenerate the cumulants. The information law is therefore not new here; what this paper adds is the score-level statement with its constant $\frac{1}{6}\|\kappa_4\|_F^2$ and its content (the fourth-order term closes the gap), the general-order constant, the hypothesis under which the expansion is a theorem and the heavy-tailed regime outside it, the crossover t_* as an exact scale, the verification of rates and constants

against an exact non-Gaussian dependent score, and the learning consequences with an exact E-step. Speciation and collapse times (Biroli et al., 2024; Achilli et al., 2026) and the information-restricted Bayesian analysis of Hunt et al. (2026) are second-moment phenomena in this vocabulary; heavy-tailed forward processes (Pandey et al., 2024; Tang et al., 2026) change the channel and therefore the ladder.

Learning from noisy data only. Ambient diffusion learns a clean distribution from corrupted samples by predicting a further corruption (Daras et al., 2023; 2024); EM with diffusion posterior sampling in the E-step (Rozet et al., 2024; Bai et al., 2024) and SURE-type surrogates (Kawar et al., 2024) address the same problem nonparametrically. Our learner is parametric and its E-step is exact, so the question changes from "can the clean law be recovered" to "how fast is it lost", which Theorem 4 answers order by order. The estimator is classical: sum-product on a chain is forward–backward (Baum et al., 1970; Rabiner, 1989), grid quadrature is Kitagawa's smoother (Kitagawa, 1987), Fisher's identity and EM are Dempster et al. (1977); Cappé et al. (2005), and the Gaussian special case is the Shumway–Stoffer algorithm (Shumway & Stoffer, 1982). Consistency of the maximum-likelihood estimator for general-state hidden Markov models is Douc et al. (2011); the identifiability of a Markov signal observed through independent noise, which is what makes the one-view protocol of Section 4 well posed, is Gassiat et al. (2020). Spectral and moment methods for hidden Markov models (Hsu et al., 2012) are the natural high-noise competitor; the moment estimators of Section 4 are their simplest form here, rate-correct and inefficient below the crossover.

Sequence denoisers and length. The baselines of Section 5 are the standard sequence architectures (Bai et al., 2018; Vaswani et al., 2017; Ronneberger et al., 2015); that attention-based length extrapolation turns on the positional encoding (Press et al., 2022; Kazemnejad et al., 2023) is visible in Table 6, here against an exact reference at every length.

7. Limitations and conclusion

The model is a chain in one dimension with a linear autoregressive form; the theorems hold for any law with an exponential moment at fixed length, and heavy-tailed laws sit outside them with a regime of their own; the exact verification and the learning results are for chains, at one driving-noise family in most experiments and with eight seeds in the regime sweeps. The estimator's advantage is the correctness of its factorisation: under a non-Markov truth it is misspecified in a way more data does not repair, and Section 5 computes where that begins. Inference costs $O(LN_g^2)$ per sequence and level—of order ten times a network forward pass at $N_g = 401$, with the mixture M-step a fixed cost per iteration that matches the E-step near $M = 2048$ and dominates below it—so the advantage is in data and understanding, not in wall-clock: a factor of eight in error at equal data is bought with a structural assumption and with compute a network does not pay. The comparison with denoisers equalises the number of clean sequences under each method's standard recipe and, separately, matches the information interface with no oracle on either side; it does not rank learning algorithms in general.

What the laboratory teaches is general: diffusion noise erases statistical structure order by order, at rates fixed by Hermite degree, so a score learner that reaches only second order is bounded away from the truth by an e^{-8t}-scaled term whose content is the fourth-order moment, and a learner of shape pays e^{4t} in samples per level. Explicit structure is the way to recover what the data still carry; the price of the wrong structure is a bias that no sample size removes, and that bias can be computed.

Impact statement

This paper presents theoretical and empirical work on how diffusion noise erases statistical structure and on learning structured dynamics from noised data, evaluated on synthetic models with known ground truth. Its societal consequences are those of a better understanding of generative diffusion models in general; we see no specific ethical concern that requires discussion here.

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A. Proofs

A.1. Proposition 1

Write $\overrightarrow{m}_i^{\mathrm{p}}$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood. We show by induction that every projected message equals the corresponding exact message of the covariance-matched Gaussian chain. The initial message is Gaussian or projected to the Gaussian with the same first two moments, which is the Gaussian chain's initial message. If $\overrightarrow{m}_{i-1}^{\mathrm{p}}$ is that chain's message, with mean m and variance v, multiplying by ℓ_{i-1} gives a Gaussian with the precision-weighted moments, identical in both chains; the transition integral against $a_i = \rho a_{i-1} + \varepsilon_i$ is the density of $\rho a_{i-1} + \varepsilon_i$, whose mean and variance are $(\rho m', \rho^2 v' + q)$ for every driving noise with moments $(0, q)$, so Π_{G} returns the Gaussian chain's next message. The backward message is $h(\rho a)$ with h the density of $a'' - \varepsilon$ under the incoming Gaussian, of mean m' and variance $v' + q$; as a function of a its normalised moments are $(m'/\rho, (v'+q)/\rho^2)$, which is where $\rho \neq 0$ is used, starting from the terminal likelihood ℓ_L (the flat terminal message is not normalisable and contributes no factor). Every projected belief is therefore the Gaussian chain's belief, whose posterior mean is linear in $\mathbf{x}$ and coincides with the LMMSE estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Since the recursion never read p_ε beyond $(0, q)$, the estimator is the same for every such law; nothing assumes stationarity, since the induction carries site-dependent moments. The algebraic steps (the two moment maps as integrals, the covariance identities, and the optimality of the normal-equation gain in arbitrary dimension) are machine-checked in Lean 4 (lean/README.md); the projection step itself is not. The two consequences stated after the proposition follow from the same fact: sum-product with a Gaussian kernel is the Gaussian chain's smoother with that kernel's covariance, a linear map, and the LMMSE of the true covariance is the best linear map in mean square; and the Gaussian log-likelihood of a noised sample is a function of its sample covariance, whose expectation $e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$ identifies (ρ, q).

A.2. Proposition 7

(i) $\mathbf{a}$ and $\mathbf{z}$ are independent and Gaussian, so $(\mathbf{a}, \mathbf{x})$ is jointly Gaussian with $\mathrm{Cov}(\mathbf{x}) = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t\mathbf{I} = \mathbf{\Sigma}_t$ and $\mathrm{Cov}(\mathbf{a}, \mathbf{x}) = e^{-t}\mathbf{\Sigma}_0$. Hence $P_t = \mathcal{N}(0, \mathbf{\Sigma}_t)$, $S = \nabla \log P_t = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$, and $\langle \mathbf{a} \rangle_{x,t} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\,\mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Then $\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t} = (\mathbf{\Sigma}_t - e^{-2t}\mathbf{\Sigma}_0)\mathbf{\Sigma}_t^{-1}\mathbf{x} = \Delta_t \mathbf{\Sigma}_t^{-1}\mathbf{x}$, which is Tweedie's identity $S = -(\mathbf{x} - e^{-t}\langle \mathbf{a} \rangle_{x,t})/\Delta_t$. (ii) Forward, by induction. $\overrightarrow{m}_1 = \mu = \mathcal{N}(0, 1)$ is the law of a_1 before any observation. If $\overrightarrow{m}_i(a) = p(a_i{=}a \mid x_{1:i-1}) = \mathcal{N}(a; m, P)$, Bayes' rule gives $\overrightarrow{m}_i \ell_i \propto p(a_i \mid x_{1:i})$, Gaussian with mean $m + k(x_i - e^{-t}m)$ and variance $(1 - ke^{-t})P$, $k = e^{-t}P/(e^{-2t}P + \Delta_t)$: the Kalman update. The Markov property then gives $\overrightarrow{m}_{i+1}(a') = \int K(a' \mid a)\overrightarrow{m}_i(a)\ell_i(a)\,\mathrm{d}a \propto p(a_{i+1}{=}a' \mid x_{1:i})$, Gaussian with mean ρ times the filtered mean and variance ρ^2 times the filtered variance plus q: the Kalman prediction. Backward, $\overleftarrow{m}_L \equiv 1$, and if $\overleftarrow{m}_{i+1}(a) \propto p(x_{i+2:L} \mid a_{i+1}{=}a)$ then $\overleftarrow{m}_i(a) = \int K(a' \mid a)\ell_{i+1}(a')\overleftarrow{m}_{i+1}(a')\,\mathrm{d}a' \propto p(x_{i+1:L} \mid a_i{=}a)$, again by the Markov property; it is the exponential of a quadratic in a (a Gaussian in information form, of zero precision at $i = L$). Since $x_{i+1:L}$ and $x_{1:i}$ are conditionally independent given a_i, $\overrightarrow{m}_i \ell_i \overleftarrow{m}_i \propto p(a_i \mid x_{1:i})\, p(x_{i+1:L} \mid a_i) \propto p(a_i \mid \mathbf{x})$. The Rauch–Tung–Striebel smoother computes the same conditional law by another factorisation, conditioning backward on a_{i+1}; two exact computations of one law agree, and its mean is $\langle a_i \rangle_{x,t}$ of (i). tests/test_kalman.py checks the three against each other to 10^{-12}.

A.3. Theorem 2

For $z \sim \mathcal{N}(0, 1)$, real a and $\alpha \in (0, 1)$, the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$ evaluated at $x = \alpha a + \sqrt{1 - \alpha^2}\, z$ and averaged over z gives $e^{s\alpha a - s^2\alpha^2/2} = \sum_n \mathrm{He}_n(a)(\alpha s)^n/n!$, hence $\mathbb{E}_z \mathrm{He}_n(\alpha a + \sqrt{1 - \alpha^2}\, z) = \alpha^n \mathrm{He}_n(a)$. The coordinates of $\mathbf{z}$ are independent given $\mathbf{a}$, so the product form holds for H_ν; integrating over $\mathbf{a} \sim P_0$ requires only $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$. The cases $n \leq 4$ are verified in Lean as polynomial identities under the first four Gaussian moments.

A.4. Theorems 4 and 5

Exact identities. With $\alpha = e^{-t}$ and $\Delta_t = 1 - \alpha^2$, Bayes' rule gives $P_t(\mathbf{x}) = \mathcal{N}_{\Delta_t}(\mathbf{x})\, \mathbb{E}_{P_0} \exp\bigl(\alpha \mathbf{x} \cdot \mathbf{a}/\Delta_t - \alpha^2 |\mathbf{a}|^2/2\Delta_t\bigr)$, where $\mathcal{N}_{\Delta_t}$ is the $\mathcal{N}(0, \Delta_t \mathbf{I})$ density. Hence $\log P_t(\mathbf{x}) = \log \mathcal{N}_{\Delta_t}(\mathbf{x}) + \Lambda(s, b)$ with $s = \alpha \mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and $\Lambda(s, b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$; $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = \nabla_s \Lambda$ and $S = -\mathbf{x}/\Delta_t + (\alpha/\Delta_t)\nabla_s \Lambda$.

Where the expansion lives. Under Assumption 3 the integrand is dominated by $e^{\lambda|\mathbf{a}|}$ on $\{|s| \le \lambda\} \times \{b \ge 0\}$, so Λ is finite there and real-analytic in s on $|s| < \lambda$ for every $b \ge 0$, with Taylor coefficients at the origin equal to the joint cumulants of the linear form $s \cdot \mathbf{a}$ and the quadratic form $|\mathbf{a}|^2/2$; the tilted laws $\propto P_0(\mathbf{a}) e^{\theta(s\cdot\mathbf{a} - b|\mathbf{a}|^2/2)}$, $\theta \in [0,1]$, have moments of every order bounded uniformly on $\{|s| \le \lambda/2,\ 0 \le b \le 1\}$ by constants depending on λ and $\mathbb{E}e^{\lambda|\mathbf{a}|}$. Without Assumption 3 the expansion at $b = 0$ is not available: $\Lambda(s,0) = \log \mathbb{E}e^{s\cdot\mathbf{a}}$ is infinite for every $s \neq 0$ when P_0 has only finitely many moments.

Grading. Expand Λ in (s,b) at the origin and grade a monomial $s^j b^k$ by $j + 2k$; since $s = O(\alpha)$ and $b = O(\alpha^2)$, the grade is the order in α. By the symmetry of P_0 under $\mathbf{a} \mapsto -\mathbf{a}$ only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$, with $\Lambda_2 = \frac{1}{2} s^\top \mathbf{\Sigma}_0 s - \frac{b}{2} \operatorname{tr} \mathbf{\Sigma}_0$ and

$$\Lambda_4 = \tfrac{1}{24}\kappa_4[s,s,s,s] - \tfrac{b}{4}\sum_i \operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i^2) + \tfrac{b^2}{8}\sum_{ij} \operatorname{cum}(a_i^2, a_j^2),$$

where cum is the joint cumulant under P_0 and the squares enter because b multiplies $|\mathbf{a}|^2/2$: $\operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i^2) = \operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i, a_i) + 2(\mathbf{\Sigma}_0 s)_i^2$ and $\operatorname{cum}(a_i^2, a_j^2) = \kappa_4[i,i,j,j] + 2(\mathbf{\Sigma}_0)_{ij}^2$, the added terms being functions of $\mathbf{\Sigma}_0$ alone. Every fourth-order joint cumulant of linear forms in $\mathbf{a}$ equals its value under the covariance twin plus a linear contraction of κ_4, so $\Lambda_4 - \Lambda_4^{\mathrm{G}}$ is linear in κ_4 and $\Lambda_2 = \Lambda_2^{\mathrm{G}}$. The remainder R_6 is the Lagrange remainder of grade six, a combination of sixth-order derivatives of Λ at an intermediate point of the segment to the origin, i.e. of tilted joint cumulants of order six; on the bulk $B = \{|s| \le \lambda/2\} = \{|\mathbf{x}| \le \lambda\Delta_t/2\alpha\}$ it is bounded by $C\alpha^6(1+|\mathbf{x}|)^6$ with C depending only on λ and $\mathbb{E}e^{\lambda|\mathbf{a}|}$, by the uniform moment bound above.

Leading term of the score gap. $S - S^{\mathrm{G}} = (\alpha/\Delta_t)\nabla_s(\Lambda - \Lambda^{\mathrm{G}}) = (\alpha/\Delta_t)\big[\nabla_s(\Lambda_4 - \Lambda_4^{\mathrm{G}}) + \nabla_s R_6 - \nabla_s R_6^{\mathrm{G}}\big]$. Substituting $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and expanding in α, the grade-four term is $\alpha^4 \nabla_{\mathbf{x}}\big[\frac{1}{24}\sum_{ijkl}\kappa_4[ijkl]\,{:}x_i x_j x_k x_l{:}\big] + O(\alpha^6(1+|\mathbf{x}|)^5)$, where $:\cdot:$ is the Wick (Hermite) product: the three contributions of Λ_4 combine into the Hermite form because $b = \alpha^2(1 + O(\alpha^2))$ and the cumulant contractions with δ_{ii} are exactly the terms that convert monomials into Hermite polynomials. The m-th component is $\alpha^4 \frac{1}{6}\sum_{jkl}\kappa_4[mjkl]\,{:}x_j x_k x_l{:}$.

Second moment. Under $\mathcal{N}$, Wick's theorem gives $\mathbb{E}_{\mathcal{N}}[{:}x_j x_k x_l{:}\,{:}x_{j'} x_{k'} x_{l'}{:}]$ equal to the number of perfect matchings of $\{j,k,l\}$ with $\{j',k',l'\}$ respecting indices; contracted with the symmetric tensor this yields $\mathbb{E}_{\mathcal{N}}\big[(\frac{1}{6}\sum_{jkl}\kappa_4[mjkl]{:}x_j x_k x_l{:})^2\big] = \frac{3!}{36}\sum_{jkl}\kappa_4[mjkl]^2$, and summing over m, $\mathbb{E}_{\mathcal{N}}\|\text{leading term}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2\alpha^8$. Replacing $\mathbb{E}_{\mathcal{N}}$ by $\mathbb{E}_{P_t}$ on the bulk costs a factor $1 + O(\alpha^2)$: $P_t/\mathcal{N} = 1 + \alpha^2\frac{1}{2}\langle \mathbf{\Sigma}_0 - \mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + O(\alpha^4(1+|\mathbf{x}|)^4)$ there, and the degree-2 correction carries an explicit factor α^2, so its contribution to the second moment is of relative size α^2. The cross term between the leading term and the grade-six remainder is $O(\alpha^{10})$ by Cauchy–Schwarz and the moment bound. The constant in $O(\alpha^2)$ depends on P_0 and L; for the Gaussian chain it is $\lambda_{\max}(\mathbf{\Sigma}_0)$ (last paragraph).

Tail. On $B^c = \{|\alpha\mathbf{x}| > \lambda\Delta_t/2\}$ the expansion is not used. By the tower property and Jensen, $\mathbb{E}_{P_t}[\|\mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|^2 \mathbf{1}_{B^c}] \le \mathbb{E}[|\mathbf{a}|^2 \mathbf{1}_{B^c}]$, and $B^c \subset \{\alpha|\mathbf{a}| > c/\alpha\} \cup \{\sqrt{\Delta_t}|\mathbf{z}| > c/\alpha\}$ with $c = \lambda\Delta_t/4$, so $P_t(B^c) \le e^{-\lambda c/\alpha^2}\mathbb{E}e^{\lambda|\mathbf{a}|} + e^{-c^2/2\alpha^2}$, which is $\exp(-c' e^{2t})$ under Assumption 3. Since $\|S - S^{\mathrm{G}}\|^2 \le C(|\mathbf{x}|^2 + \alpha^2\|\mathbb{E}[\mathbf{a} \mid \mathbf{x}]\|^2 + \alpha^2\|\mathbb{E}^{\mathrm{G}}[\mathbf{a} \mid \mathbf{x}]\|^2)/\Delta_t^2$, Cauchy–Schwarz against $P_t(B^c)^{1/2}$ and the finiteness of $\mathbb{E}|\mathbf{a}|^4$ make the tail contribution below every power of α. Together, $\mathbb{E}_{P_t}\|S - S^{\mathrm{G}}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}(1 + O(e^{-2t}))$, which is Equation (3). Without Assumption 3 the tail probability is polynomial in α (of order $\alpha^{2\nu}$ for a tail index ν), and for $\nu \le 8$ the tail's own contribution is of order α^ν and leads; this is the heavy-tailed regime of Section 3.

Information and the general order. $\partial_\theta \log P_t = \partial_\theta \Lambda$. If $\partial_\theta \mathbf{\Sigma}_0 \neq 0$ the grade-two term $\frac{\alpha^2}{2}\langle \partial_\theta \mathbf{\Sigma}_0, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle$ leads, and $\mathbb{E}_{\mathcal{N}}$ of its square is $\frac{\alpha^4}{4} \cdot 2\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$ by Wick, giving Equation (4). If $\partial_\theta \mathbf{\Sigma}_0 = 0$ the grade-four term $\frac{\alpha^4}{24}\sum \partial_\theta \kappa_4[ijkl]{:}x_i x_j x_k x_l{:}$ leads, with $\mathcal{N}$-second moment $\frac{4!}{24^2}\|\partial_\theta \kappa_4\|_F^2 \alpha^8$, giving Equation (5); the corrections and the tail are handled as above. For Theorem 5, the graded expansion of Λ and the expansion $\log\big(1 + \sum_{|\nu|\ge 2}\alpha^{|\nu|}\mathbb{E}_{P_0}[H_\nu]H_\nu(\mathbf{x})/\nu!\big)$ obtained from Mehler's kernel are two formal series in α for the same function whose coefficients at each grade are polynomials in the moments of P_0 up to that grade; they agree on every compactly supported P_0, where both converge, hence identically. The grade-m coefficient of $\partial_\theta \log(P_t/\mathcal{N})$ is therefore $\alpha^m \sum_{|\nu|=m} \partial_\theta \mathbb{E}[H_\nu]H_\nu(\mathbf{x})/\nu!$ plus ∂_θ of products of lower-grade terms, each of which vanishes by the

minimality of m; its $\mathcal{N}$-second moment is $\alpha^{2m}\sum_{|\nu|=m}(\partial_\theta \mathbb{E}H_\nu)^2/\nu!$ by orthogonality, and the bulk correction and the tail are as before. Since $:\mathbf{x}\mathbf{x}^\top:$ and the degree-four Hermite polynomials are $\mathcal{N}$-orthogonal, the leading cross-information between a degree-2 and a degree-4 coordinate vanishes, so the leading term of the efficient information of the degree-4 coordinate after degree-2 nuisances equals that of its raw information. For the Gaussian chain $\mathbf{x}_t \sim \mathcal{N}(0, \mathbf{\Sigma}_t(\theta))$ exactly and the stated trace formula is the Gaussian Fisher information; its limit $e^{4t} I_t \to \frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2$ follows from $\mathbf{\Sigma}_t \to \mathbf{I}$ and $\partial_\theta \mathbf{\Sigma}_t = \alpha^2 \partial_\theta \mathbf{\Sigma}_0$, and is machine-checked in Lean (lean/README.md, target 3).

Corollary 6. At fixed $t > 0$ the family $\{P_t^\theta\}$ is differentiable in quadratic mean with finite, non-singular information (smoothness of Λ in θ under Assumption 3), so the local asymptotic minimax theorem of Hájek and Le Cam applies and the maximum-likelihood estimator is asymptotically efficient; the stated bound is $1/I_t^{\mathrm{eff}}$ with the efficient information of Theorem 5.

The crossover. For the Gaussian chain the expansion of $P_t/\mathcal{N}$ is an expansion in $\alpha^2\mathbf{\Sigma}_0$, i.e. in $\alpha^2\lambda_{\max}(\mathbf{\Sigma}_0)$ in operator norm, with $\lambda_{\max} \le (1+\rho)/(1-\rho)$ by the Toeplitz bound $\lambda_{\max} \le \sum_k \rho^{|k|}$; $\mathbf{\Sigma}_t^{-1}$ is within $\alpha^2\lambda_{\max}$ of $\mathbf{I}$ in operator norm, which is the source of $\epsilon_t = O(e^{-2t}\lambda_{\max})$ and of $t_* = \frac{1}{2}\log\lambda_{\max}$. The same chain has $P_t/\mathcal{N} \in L^2(\mathcal{N})$ iff $2\mathbf{I} - \mathbf{\Sigma}_t \succ 0$, i.e. iff $e^{2t} > \lambda_{\max} - 1$, so the Hermite expansion of $P_t/\mathcal{N}$ converges in $L^2(\mathcal{N})$ exactly above that time; for the Laplace chain it never does, which is why the proof goes through Λ.

B. Numerical representation and its control

The recursion. The two message sweeps of Section 2 are

$$\begin{aligned} \overrightarrow{m}_i(a) &\propto \int K(a \mid a')\, \ell_t(x_{i-1} \mid a')\, \overrightarrow{m}_{i-1}(a')\, \mathrm{d}a', \\ \overleftarrow{m}_i(a) &\propto \int K(a'' \mid a)\, \ell_t(x_{i+1} \mid a'')\, \overleftarrow{m}_{i+1}(a'')\, \mathrm{d}a'', \end{aligned} \tag{7}$$

with $\overrightarrow{m}_1 \equiv \mu$, the initial law, and $\overleftarrow{m}_L \equiv 1$; the site marginal is $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ and Tweedie's identity turns its mean into the score.

Messages are represented on a uniform grid of N_{g} points on $[-A, A]$ with composite trapezoidal weights; the recursion is a scaled linear-domain forward–backward whose discarded constants are accumulated so the evidence $\log P_t(\mathbf{x})$ of the discretised model is exact to floating point. Three levels of exactness are kept apart: sum-product is exact at the level of functional messages; forward–backward is exact for the discretised factor model; and that model approximates the continuous chain with truncation (mass outside $[-A, A]$) and quadrature (second order in the spacing) errors that are measured, not assumed.

Validation where the answer is known.

Proposition 7 (The Gaussian chain: score, smoother and sum-product coincide). Let the driving noise of Equation (1) be Gaussian, so that $\mathbf{a} \sim \mathcal{N}(0, \mathbf{\Sigma}_0)$ with $(\mathbf{\Sigma}_0)_{ij} = \rho^{|i-j|}$, and let $\mathbf{x}$ follow Equation (2). Then (i) $S(\mathbf{x}, t) = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ and $\langle \mathbf{a} \rangle_{x,t} = e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$, with $\mathbf{\Sigma}_t = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$, and Tweedie's identity holds between them; (ii) every message of Equation (7) is Gaussian, $\overrightarrow{m}_i$ is the Kalman filter's prediction density $p(a_i \mid x_{1:i-1})$ and $\overrightarrow{m}_i \ell_t$ its filtering density, and the site marginals $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ are those of the Rauch–Tung–Striebel smoother, whose means are therefore $\langle \mathbf{a} \rangle_{x,t}$ of (i).

The proof is in Appendix A; Figure 5 measures it (e39). An RTS smoother written without the BP code (src/markov_score/kalman.py) agrees with the closed form to 1.7×10^{-15}, and grid BP at the operating grid agrees with both to 1.9×10^{-13} at every one of the 12 levels. Refining the grid converges faster than any power of the spacing, as the trapezoid rule does on Gaussian integrands, and reaches 10^{-10} by $N_{\mathrm{g}} = 81$ at the smallest level, where the likelihood is narrowest, and by $N_{\mathrm{g}} = 41$ at the largest; widening the domain at the operating spacing removes the truncation error, which is 9.9×10^{-2} at $A = 3$. On the same grid the moment-projected recursion of Proposition 1 is off by up to 2.6×10^{-1}: its backward messages are Gaussians whose variance grows like e^{2t}, and the grid truncates them at $\pm A$. The proposition concerns functional messages, and the LMMSE floors reported

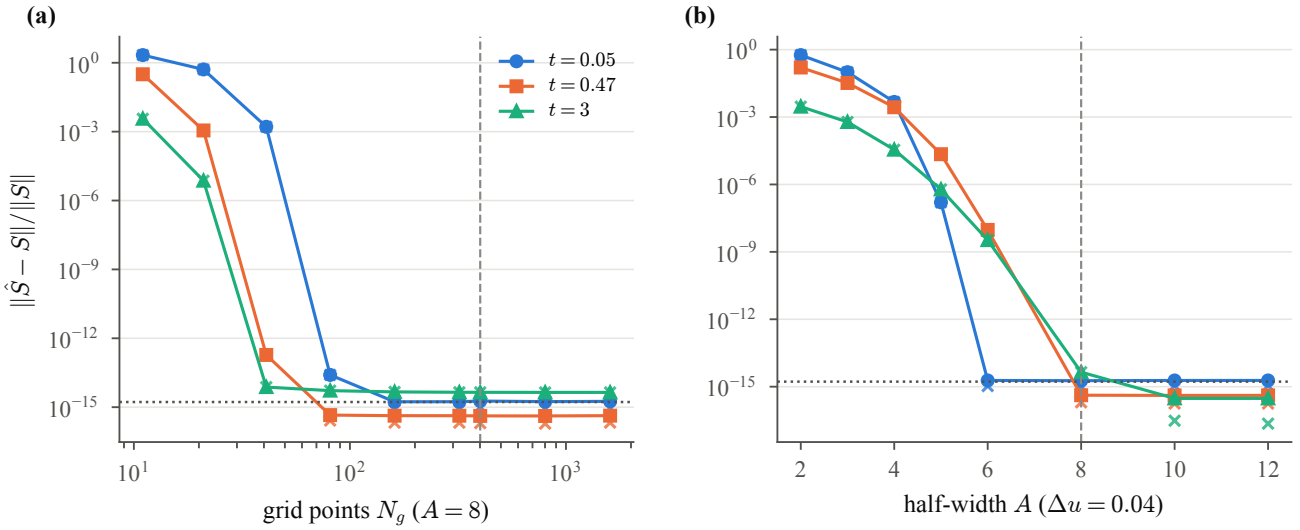


Figure 5. The Gaussian sanity check (e39). Pooled relative score error of grid BP against the closed-form score (lines) and against an independent Kalman–Rauch–Tung–Striebel smoother (crosses) on the Gaussian chain, at the smallest, a middle and the largest level. (a) Refining the grid at $A = 8$; (b) widening the domain at the operating spacing. Dashed: the operating grid. Dotted: the smoother against the closed form.

here are computed in closed form. Grid BP also reproduces brute-force enumeration of the discretised posterior on short chains to double precision (tests/test_gaussian_bp_equivalence.py, tests/test_fisher_bruteforce.py).

Against a finer reference ($N_g = 1601$, $A = 12$) the worst relative score error over the twelve levels and four driving-noise families on ordinary draws is $9.9e-04$; under a three-standard-deviation displacement the heavy-tailed families lose two orders (1.7e-02). The worst forward-message boundary occupancy is $6.6e-05$ and the worst belief edge occupancy $2.9e-03$ over 60 production cells.

The envelope, measured rather than inherited. The validation above fixes the grid at the frozen configuration. Every other configuration this paper touches—the other driving-noise families, the correlations of the family ladder, the longer chains of the length study—inherits it, so we measured it instead: 5 driving-noise families × 4 correlations × twelve levels × 4 seeds, seven candidate grids against a reference at $N_g = 3201$, $A = 12$.

At the frozen configuration the operating grid is comfortable: the pooled drift of the posterior-mean field is 9.2×10^{-6}, the worst single level 3.7×10^{-5}, and 2.8×10^{-4} of the belief mass lies in the outermost 2% of the domain. Nothing reported at that configuration is in question.

Over the whole sweep the picture is different, and the difference is instructive: at half-width 8 the worst-case drift is flat in the number of grid points, 5.4×10^{-2} at $N_g = 201$ and 5.4×10^{-2} at $N_g = 1601$, and falls only when the domain widens—2.6×10^{-2} at $(401, 10)$ and 3.0×10^{-3} at $(801, 12)$. That is the signature of truncation rather than of quadrature, and it means refinement in N_g alone, which is what refinement has meant in this literature, cannot fix it. The binding cell is the *student* family at $\rho = 0$, and the mechanism is a handful of sites: among the 8192 sites of each seed's bundle, $1, 1, 1, 2$ lie beyond $|a| = 8$ (the analytic expectation is 1.2), the largest at $|a| = 15.4$. A site beyond the domain has its posterior pinned to the boundary on every grid, which is the "belief mass 1.00 in the outer 2%" of the diagnostic, and at the lowest level its score error alone is the drift of 5.4×10^{-2}. For a polynomial tail the expected exceedance count is Poisson with mean $MLP(|a| > A)$, which no finite domain makes zero: 0.17 per 8192 sites at $A = 12$, 0.04 at $A = 16$. Per replicate of the family ladder of Appendix E (512 chains of 32 sites) the same expectation is 2.39 sites at $A = 8$, 0.307 at 12 and 0.0655 at 16 for the Student-t_5 chain at $\rho = 0$, and below 0.188 for every other family and correlation; that ladder therefore quotes the heavy-tailed family from a recomputation on the $(1601, 16)$ domain, where its pooled floor has stopped moving (e07b, e07c); no claim in the body rests on it. The same consideration governs the closure panel of Figure 2: beyond $t \approx 3$ the reference on the $(801, 8)$ grid carries the tail of the truncated prior, and the heavy-tailed families depart from the constant for that reason alone; on $(3201, 32)$ they reach it (Appendix K).

Table 3. Architecture screen under one recipe on the development seeds (never used for a reported number): the best configuration per family, its parameter count, its pooled bulk risk over 4 development seeds, two sizes and 3 restarts of 32,000 steps per configuration, and the risk of the family's runner-up configuration.

architecture	selected configuration	parameters	screen risk	runner-up
MLP (global)	hidden: (128, 128), lr 0.001, eps	26,016	0.390	0.411
window head	radius: 4, width: 64, lr 0.001, eps	5,505	0.116	0.122
dilated TCN	channels: 32, dilations: (1, 2, 4, 8), lr 0.001, eps	17,057	0.143	0.160
bidirectional GRU	hidden: 32, lr 0.001, eps	15,169	0.127	0.133
transformer	d: 32, heads: 4, layers: 2, lr 0.003, eps	18,057	0.147	0.154
1-D U-Net	channels: 16, lr 0.001, eps	13,713	0.148	0.212

The envelope also has to cover the longer chains of the length study and of the regime comparison, and it does (e11b: the same sweep at $L \in \{64, 128\}$, 4 seeds). On the operating grid the worst pooled drift over the sweep is 5.4×10^{-2} at $L = 32$, 5.8×10^{-2} at 64 and 6.6×10^{-2} at 128, a factor of at most 1.22, and at the frozen configuration 9.2×10^{-6}, 7.2×10^{-6} and 4.5×10^{-6}: the per-site quadrature error does not compound along the chain, because the messages are renormalised at every site.

A second observation from the same sweep cuts the other way. The column-mass defect of the discretised kernel, $\max_j |\sum_k K(u_k \mid u_j) w_k - 1|$, grows with correlation, from 7.0×10^{-3} at $\rho = 0$ to 0.15 at $\rho = 0.95$, and the reason is again the domain and not the spacing: the maximum sits in the outermost columns, where a kernel centred at ρu_j with $|u_j|$ near A loses the part of its mass beyond the domain, so the defect is flat in N_{g}, falls as the domain widens, and is at machine precision in the interior columns at every ρ. It is not predictive of the error: at $\rho = 0.95$ the drift is 1.2×10^{-2}, because the recursion renormalises every message and the affected columns carry almost no belief. We report it as a diagnostic and not as an error bar. The uniform family is the one exception to the quadrature picture: its kernel is discontinuous, the trapezoidal rule is first-order on it, and its worst-level drift is largest on the coarsest spacing rather than the narrowest domain.

Refinement on the reported estimands. Re-scoring the validation-selected kernels of the capacity experiment on their own test bundles at $N_{\mathrm{g}} = 801$ and 1601 ($A = 8$) and at $A = 10$ ($N_{\mathrm{g}} = 801$), with the reference recomputed at each resolution, moves the pooled error of EM–BP ($C = 8$) by at most 0.00%, 0.00% and 0.00% at $M = 2048$ (maximum over 16 seeds), and the LMMSE floor by at most 0.00%; the reference score itself moves by 0.001% between $N_{\mathrm{g}} = 401$ and 801. The reported effects are three orders of magnitude larger than the discretisation.

Floors in the theory verification. The outer-product information estimate of a shape coordinate is a small residual of order-one quantities; for the kinked Laplace density its quadrature floor is reached near $t = 2.5$ at $N_{\mathrm{g}} = 801$ (the replicate spread stops shrinking), which is why Figure 2b reports the asymptotic slope on the smooth members of the family. The closure gap of the uniform family reaches the double-precision floor of the BP means near $t = 3.5$, where its squared gap is 10^{-13} per chain.

C. Experimental protocol and provenance

Frozen configuration. Laplace chain unless stated, $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0, 1)$, $L = 32$; grid $N_{\mathrm{g}} = 401$, $A = 8$; twelve log-spaced levels $t \in [0.05, 3.0]$; other families Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$), generalised Gaussian ($\beta \in [1, 6]$), all variance-matched. Sixteen confirmatory seeds 0–15 for every comparison at the frozen configuration; eight for the regime sweeps; development seeds 10000–10003 for every architecture and hyperparameter choice (three restarts of 32,000 steps per configuration at $M \in \{32, 2048\}$, ranked by pooled bulk risk; Table 3). Validation and test bundles are drawn per seed (1024 and 2048 sequences) from streams disjoint from every training stream; their hashes are stored with every row so pairing across arms is checkable.

Estimand. Schedule-pooled relative score error $\left(\sum_t \sum_{j,i} (\hat{s}_t(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i)^2 / \sum_t \sum_{j,i} s_t(\mathbf{x}_j)_i^2\right)^{1/2}$ on all sites (the headline), on the interior sites 5–28 (used for selection), and on the centre site (diagnostic). Ratios are formed within seed and averaged; intervals are percentile bootstraps over seeds.

Arms and interfaces. EM–BP: mixture kernel, C as stated, ρ initialised at 0.3, one noised view per sequence with sequences split evenly across the twelve levels (interface C), checkpoint ladder $\{10, 20, 40, 60, 80, 120, 160, 220, 300, 400, 600, 800, 1200\}$ selected on the validation bundle scored against the true kernel; selecting instead on held-out one-view log-evidence, which needs no true score, gives the same test error within seed noise (Appendix K). Networks under interface A: trained by denoising score matching on the clean sequences with the level and the corruption redrawn at every step, batch 64, Adam, gradient clipping at 1, both output parameterisations, checkpoint ladder to 32,000 steps selected on the validation bundle. Interface B: the same objective on one fixed corruption per clean sequence, the checkpoint selected as under interface A on the validation bundle scored against the true kernel. Interface C: implicit score matching on the same one-view noisy bundle as EM–BP, with the divergence computed exactly for the window head and by Hutchinson's estimator otherwise, and the checkpoint selected on the implicit objective of a held-out noisy bundle; the true score enters only in the final scoring. The misspecification experiment's network is a radius-2, width-64 window head trained for 16,000 steps under interface A.

Provenance. Every run directory carries the commit, the dirty-tree list on the code path, the resolved configuration digest, the seeds, host and environment; a generator refuses to emit a table from a run whose code path was dirty or from an incomplete design, and flags, without dropping, any cell whose narrowest fitted component is under two grid cells. Every number in this paper is a macro emitted by such a generator; the manuscript check (tools/check_manuscript.py) fails the build on any typed decimal, and an independent re-derivation of every headline number from the frozen files (tools/audit_claims.py) runs with it. The regime sweep was recomputed at twice the seed count in a fresh output directory, from a different deployment of the source and on a different compute node; across the 480 rows it shares with E13, every risk and every fitted moment agrees to the last bit (maximum absolute difference 0 in the pooled error and 0 in the fitted excess kurtosis).

D. The oracle ladder

Table 4. Schedule-pooled relative score error of the two oracles against the tree-exact reference, 512 chains per family at the frozen configuration. The LMMSE column is the floor of every second-order estimator (Proposition 1).

family	excess kurtosis	isotropic $-\mathbf{x}$	LMMSE, true $\mathbf{\Sigma}_0$
Laplace	$+3$	0.670	0.168
Student-t_5	$+6$	0.668	0.158
uniform	-1.2	0.671	0.149
bimodal	-0.72	0.663	0.082
Gaussian	0	0.662	0.000

E. The second-order floor across correlation

The frozen configuration fixes $\rho = 0.85$. The price of second-order closure is a function of ρ as well as of the driving-noise law, and this appendix reports it over $\rho \in \{0, 0.3, 0.6, 0.85, 0.95\}$ for five families on the twelve-level schedule, computed from the exact score with no estimation involved. The heavy-tailed family is quoted from a recomputation on the $(1601, 16)$ domain (e07c): on the operating grid the quadrature sweep of Appendix B places its $\rho = 0$ cell outside the validated envelope, one to two sites per bundle lying beyond the domain and pinned to its boundary. Widening the domain moves its pooled floor by at most 0.005 from $(801, 8)$ to $(801, 12)$ and by at most 0.0002 from $(801, 12)$ to $(1601, 16)$ (e07b, e07c), and every other family by at most 0.0010 between the operating and the widest grid; the other families are quoted from the operating grid, where they are converged.

Across correlation. The floor is a function of the driving-noise law and of ρ (Figure 6): on the Laplace chain it is 0.295 for independent sites, 0.168 at the frozen $\rho = 0.85$ and 0.085 at $\rho = 0.95$, where $t_* = 1.50$ and the covariance carries most of the score energy until late in the schedule. The families order by the size of their fourth cumulant at every ρ; the isotropic reference rises with ρ because the covariance itself is what an unstructured estimator must learn first.

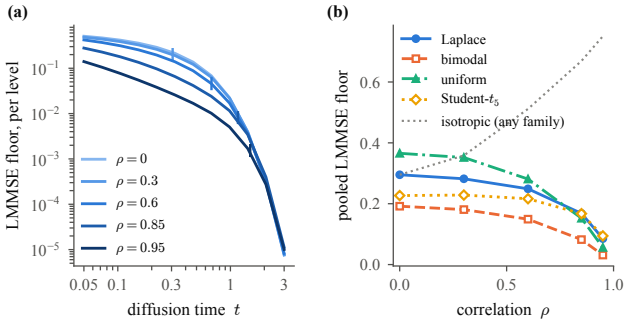


Figure 6. The second-order floor across correlation. (a) Per-level LMMSE floor on the Laplace chain for five ρ (ticks: $t_*(\rho)$); the floor collapses after the crossover. (b) Schedule-pooled floor against ρ for four driving-noise families, with the isotropic reference: strong correlation makes the score energy second-order-dominated, so the relative price of closure falls with ρ while the absolute fourth-order structure does not. The Student-t_5 curve is computed on the $(1601, 16)$ domain, where its $\rho = 0$ cell is converged (Appendices B and E); the other families on the operating grid.

F. Capacity: the previous reading, and why it is void

An earlier study of this model compared mixture kernels of $C \in \{1, 2, 4, 8, 16\}$ at $M \in \{128, 512\}$ over sixteen seeds and concluded that no capacity improved on a single linear-Gaussian kernel, attributing the estimator's margin over the networks to the Markov factorisation and its covariance alone. That sweep is void for a reason prior to any reading of it: its training draw re-created the random generator for every sequence, so every arm at every capacity and seed was fitted to a single chain of L sites (its validation and test bundles were drawn correctly). The signature is in its own numbers: Proposition 1 places a correctly fitted $C = 1$ kernel on the LMMSE floor, 0.168 on the Laplace chain, where Section 5 and all fifteen cells of the regime sweep land to 0.0099, and the earlier sweep's single-Gaussian arm scatters across a range several times wider. The stopping asymmetry that sweep also shows (its likelihood tolerance fired for almost every single-Gaussian cell and for no cell with $C \geq 4$, so that the richer arms reported checkpoints far past their optimum) is the behaviour of em on one chain, not evidence about capacity in either direction; we had first read it as the explanation of the earlier conclusion, and withdraw that reading. The capacity experiment of Section 5 draws every chain independently, asserts it, and selects the checkpoint identically for every capacity on a per-seed validation bundle; the same defect was found and fixed in one of our own drivers before any of its numbers reached a page (Appendix C).

G. Capacity across the driving-noise family

Moving the driving noise from Laplace (excess kurtosis +3) through the Gaussian to the platykurtic end at fixed ρ and variance (Figure 7), the LMMSE floor traces a V: 0.166 at the Laplace end, exactly zero at $\beta = 2$ where the covariance twin is the truth, and 0.119 at the platykurtic end. The linear-Gaussian kernel follows that V within its own standard error at every member, which is Proposition 1 read across a family rather than at a point; the mixture kernel stays between 0.017 and 0.035 throughout. The paired gain of $C = 8$ over $C = 1$ is therefore ordered by κ_4: 0.131 ($[0.127, 0.135]$) at Laplace, 0.075 at $\kappa_4 = +1.53$, and 0.090 at the platykurtic end. At the Gaussian member the gain reverses: -0.008 ($[-0.011, -0.005]$), a small but unambiguous loss, because there is no fourth-order structure to buy and the extra components only add estimation variance. Non-Gaussian capacity is worth what Theorem 4 says it is worth—the fourth-order structure the covariance twin leaves behind—and nothing where the twin leaves nothing.

H. The capacity contrast across regimes

The capacity result above is measured at one point of the model's parameter space, so we re-ran the contrast that carries it across 15 regimes: correlations from 0.3 to 0.95 and chain lengths from 16 to 64, everything else at the frozen protocol, 16 seeds each. The mixture beats the Gaussian kernel in *all* cells with the paired interval excluding zero, by -0.253 at the most favourable corner and -0.047 at the least. The sweep also checks the apparatus at every point rather than only at the frozen configuration: Proposition 1 says the $C = 1$ arm must coincide with the LMMSE of the true covariance, and it does, to 0.0099 in the worst of the 15 cells, with a fitted

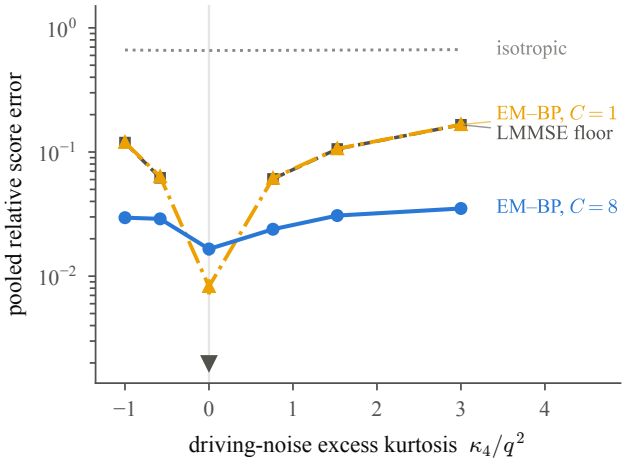


Figure 7. Capacity buys exactly what the closure discards. Pooled relative score error against the driving noise's excess kurtosis for the generalised-Gaussian family at fixed ρ and variance ($M = 512$, 16 seeds, $\pm$ s.e.): the linear-Gaussian kernel tracks the LMMSE floor, which is the exact price of second-order closure for each member; the mixture kernel tracks the exact score, and the two meet at the Gaussian member, where there is nothing beyond second order to learn. At that member the floor is exactly zero—the covariance twin is the truth—which a log axis cannot show; the line is broken there and the caret marks the point.

excess kurtosis of at most 0.000 in absolute value. The $C = 8$ arm recovers the driving noise's true excess kurtosis of 3 everywhere (2.92 to 3.31 across correlations), and its spread widens from ± 0.33 to ± 1.27 as ρ rises—the shape becomes harder to see, not differently placed.

The advantage itself shrinks monotonically with ρ, which is the direction Section 3 gives: a strongly correlated chain has a nearly Gaussian marginal, the stationary marginal's excess kurtosis falling as $3(1-\rho^2)/(1+\rho^2)$ from 2.50 at $\rho = 0.3$ to 0.15 at $\rho = 0.95$, so there is less fourth-order structure for capacity to buy. We report this as a direction and not as a rate: the pooled error runs over a schedule whose informative window moves with ρ, so its trend is not the exponent of Theorem 4 and we do not fit one to it.

I. Generation and length extrapolation: the tables

J. The matched comparison across regimes

The arms and selection rules are those of Section 5 with two additions: the evidence-selected EM–BP row, which needs no true score (Appendix K), and the network-favourable interface B row. A network cell is excluded from the factor only if its selected held-out implicit objective is not below that of the trivial score $-\mathbf{x}$ on the same views, which is what a fit that diverged looks like on its own criterion; no cell is excluded (0 network cells fail the test), and 6 carry the curve-shape flag. The two EM–BP rows agree: the same rung is selected in 52 of 96 cells and the largest difference in test error is 0.019; the last rung of the ladder is selected in 68 of 96 oracle-selected cells and 56 of 96 evidence-selected ones, so the EM–BP error is an upper bound there and every factor is conservative. Against the second-order floor, tested per seed on the same test bundle, the best network sits below it in $(0.5, 32), (0.5, 128), (0.85, 128)$ and above it in $(0.85, 32), (0.95, 32), (0.95, 128)$. At the frozen point it is above by $+0.003$ [$+0.000$, $+0.005$], a ratio of 1.02: at matched information the best network reaches the floor there and does not pass it, where with clean sequences and a fresh corruption at every step 5 of six architectures pass it (Section 5). At $\rho = 0.95$ the margin is $+0.110$ against exact radius-4 and radius-8 floors of 0.072 and 0.025 (Appendix K): the miss is the architecture's, not the window's. At the frozen point the factor is 8.3 with per-seed factors from 4.1 to 19.9, against 7.9 with per-seed factors from 5.1 to 14.6 over the sixteen seeds of Section 5: the same population at a different seed count. The networks' error falls with L because each noised view carries proportionally more sites; so does EM–BP's, and the factor survives.

Platykurtic driving noise (e14b). Every matched comparison above uses the Laplace driving noise, whose excess kurtosis is positive, so the advantage could in principle be a statement about heavy tails. The frozen cell repeated

Table 5. Generated samples against the true law at $t_{\min}$: mean log-likelihood gap $\Delta\ell$ to the reference sample under the true $P_{t_{\min}}$ (exact evidence; the exact arm's gap is the sampler's discretisation floor), driving-noise excess kurtosis $\hat{\kappa}_4$, lag fourth cumulant, sliced W_2; the row marked reference carries the reference sample's own statistics. Mean over seeds.

M	arm	$\Delta\ell$	$\hat{\kappa}_4$	lag-κ_4	W_2
128	reference	–	1.94	0.000	–
	exact	0.074	1.86	0.000	0.051
	em–bp	0.204	1.68	0.000	0.078
	lmmse	1.761	0.00	0.000	0.053
	isotropic	46.241	0.00	0.006	0.244
	net:bigru	0.752	1.10	0.006	0.160
	net:window	1.284	1.27	0.004	0.154
2048	reference	–	1.94	0.000	–
	exact	0.070	1.87	0.000	0.051
	em–bp	0.164	1.80	0.001	0.055
	lmmse	1.766	0.00	0.000	0.053
	isotropic	46.295	0.00	0.010	0.244
	net:bigru	0.385	1.55	0.003	0.167
	net:window	1.115	1.46	0.003	0.139

Table 6. Length extrapolation at $M = 2048$: pooled relative score error of estimators fitted at $L = 32$ and evaluated against the tree-exact score of longer chains, mean over 16 seeds. Longer chains are slightly easier (proportionally fewer boundary sites), so a length-agnostic arm should improve a little; the transformer instead breaks down beyond its training length.

arm	$L = 32$	$L = 64$	$L = 128$	$L = 256$
bi-GRU	0.060	0.057	0.057	0.056
EM–BP	0.023	0.023	0.023	0.023
TCN	0.068	0.066	0.065	0.064
transf.	0.063	0.243	0.514	0.703
U-Net	0.066	0.064	0.062	0.062
window	0.118	0.099	0.088	0.082

with the symmetric two-component mixture, whose excess kurtosis is negative at the same covariance, answers it: the second-order floor is 0.082, well below the Laplace value, because there is less fourth-order structure for the closure to discard; the best network, here the transformer, reaches 0.159, close to its Laplace error, and therefore sits a factor 1.93 above the floor rather than at it; EM–BP reaches 0.021, a paired factor 8.8 [7.4, 10.2] in 16 of 16 seeds, a factor indistinguishable from the Laplace factor at the same point. The estimator's margin is the fourth-order structure the covariance twin leaves behind, of either sign.

K. Reference computations behind the body

Six deterministic or cheap computations support statements in the body; each is a registered experiment (e15–e21) whose outputs are frozen like every other.

The locality floor (e15). For the Laplace chain at $\rho \in \{0.5, 0.85, 0.95\}$ and every level of the schedule, the best estimator of a_i from the window $x_{i-r}, \ldots, x_{i+r}$ is its posterior mean given the window, computed exactly by sum-product on the sub-chain with the exact marginal law of the window's first site (propagated on the grid) as initial factor; its relative score error against the full-chain score is a floor for every radius-r denoiser. At $\rho = 0.85$ the pooled floors are 0.666, 0.094, 0.030, 0.007 and 0.001 at $r = 0, 2, 4, 8, 16$; the Gaussian twin's windowed LMMSE is 0.173 at $r = 4$ against its full value 0.170, so the non-Gaussian part of the score is what a wider window carries. At $\rho = 0.5$ the radius-4 floor is 0.002. For the Gaussian chain the score's dependence on a site at distance k decays as $r(\rho, t)^k$ with $r = (A - \sqrt{A^2 - B^2})/B$, $A = \Delta_t(1 + \rho^2) + e^{-2t}(1 - \rho^2)$, $B = 2\Delta_t\rho$, from 0 at $t = 0$ to ρ as $t \to \infty$.

Table 7. Interface C across correlation and length (e14): pooled relative score error of the LMMSE oracle, of EM–BP with its iteration count chosen by held-out one-view log-evidence (ev.) and by validation risk against the true kernel (bp), of the best network trained by implicit score matching on the same one-view bundle (C) and of the best network given the clean sequence with one fixed corruption (B), each re-screened per regime on two development seeds; last column the paired factor of the C network over the evidence-selected EM–BP with a percentile bootstrap over 16 seeds. $^{\dagger n}$: n cells whose held-out objective never improved on its first checkpoint (reported, not excluded).

ρ	L	lmmse	EM–BP (ev.)	EM–BP (bp)	best net (C)	best net (B)	factor C / EM–BP (ev.)
0.5	32	0.265	0.026	0.024	0.137	0.129	5.7 [4.9, 6.7]
0.5	128	0.268	0.020	0.018	0.089	0.079	4.6 [4.1, 5.0]
0.85	32	0.167	0.024	0.021	0.170	0.156	8.3 [6.6, 10.3]
0.85	128	0.168	0.015	0.015	0.106	0.095	8.0 [6.7, 9.2]
0.95	32	0.086	0.021	0.019	0.196	0.128	10.3 [8.8, 11.9]$^{\dagger 6}$
0.95	128	0.087	0.017	0.016	0.105	0.094	6.4 [6.0, 6.9]

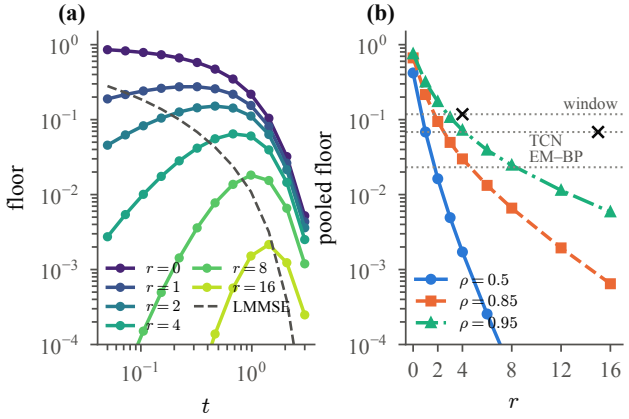


Figure 8. Locality against its floor. (a) Relative score error of the exact radius-r posterior mean on the Laplace chain at $\rho = 0.85$ at every level of the schedule, with the Gaussian twin's full LMMSE for scale: at low noise a window of one site is nearly enough, at $t \approx 1$ a radius of 16 still costs a thousandth. (b) The floor pooled over the schedule against the radius for the three sweeps; dotted lines mark the frozen all-site errors at $M = 2048$ of the radius-4 window head (the screened winner), the dilated TCN (receptive radius 15) and EM–BP, the crosses their radii. The window head sits four times above its floor; the TCN sits well above the floor of any radius past 8.

The fourth-order correction (e16). With $y = \mathbf{\Sigma}_t^{-1}\mathbf{x}$, $K = e^{-4t}\kappa_4$ and $A_{ml} = \sum_{jk} K_{mjkl}(\mathbf{\Sigma}_t^{-1})_{jk}$, the first Edgeworth correction around the twin is $\epsilon(\mathbf{x}) = \frac{1}{24}\big[K[y,y,y,y] - 6\,y^\top A y + 3\langle A, \mathbf{\Sigma}_t^{-1}\rangle\big]$ and the corrected score $S^{\mathrm{G}} + \nabla\epsilon/(1+\epsilon)$ with $\nabla\epsilon = \frac{1}{6}\mathbf{\Sigma}_t^{-1}K[\cdot,y,y,y] - \frac{1}{2}\mathbf{\Sigma}_t^{-1}Ay$. Fraction of the squared closure gap closed on the Laplace chain: 0.300 at $t = 0.22$, 0.826 at $t = 0.32$, 0.961 at $t = 0.68$, 0.985 at $t = 0.98$, 1.000 at $t = 2.07$; the bimodal and uniform chains cross ninety per cent at $t = 0.47$ and 0.47. Pooled over the schedule the corrected score is worse than the twin, because the lowest levels dominate the pooled energy and the expansion diverges there: the correction is a statement about what the gap is made of at each level, not a competitor on the schedule.

Closed-form misspecification bias (e17). For a Gaussian truth with covariance $\mathbf{\Sigma}(\delta)$ the population limit of the linear-Gaussian fit under the one-view protocol is $\theta^*(\delta) = \arg\min_{(\rho,q)} \sum_t \mathrm{KL}(\mathcal{N}(0, \mathbf{\Sigma}_t(\delta)) \,\|\, \mathcal{N}\big(0, \hat{\mathbf{\Sigma}}_t(\rho,q)\big))$ and its pooled relative score error is $b(\delta)^2 = \sum_t \mathrm{tr}[(\hat{\mathbf{\Sigma}}_t^{-1} - \mathbf{\Sigma}_t^{-1})\mathbf{\Sigma}_t(\hat{\mathbf{\Sigma}}_t^{-1} - \mathbf{\Sigma}_t^{-1})]/\sum_t \mathrm{tr}\,\mathbf{\Sigma}_t^{-1}$. Against the frozen EM–BP errors at $M = 2048$: the second-lag coefficient at the largest strength 0.247 (KL projection), 0.216 (lag-one plug-in) versus 0.214 measured; site-varying ρ at its largest strength 0.268 versus 0.287; long-range coupling 0.235, 0.202 versus 0.204; rank-one latent at $\delta = 1$ 0.065 versus 0.052. The largest deviation of the lag-one projection from the measured error over all cells is 0.019. Small-δ laws: $b \approx 0.60\,\delta$ (second lag), $1.75\,\delta$ (site-varying), $1.07\,\delta$ (long-range, local exponent 0.79), $0.15\,\delta^2$ (rank-one).

The closure constant and the heavy-tailed regime (e18). The per-sequence squared gap is a degree-six polynomial in nearly Gaussian variables and heavy-tailed (maximum over 8192 sequences 283 times the median): the relative

standard error of the mean is 4% from 8192 sequences, hence about four times that from 512. The constant is therefore quoted from 8192 sequences with a bootstrap interval: 0.92 $[0.84, 1.01]$ at $t = 3$ and 0.99 $[0.91, 1.08]$ at $t = 4$ for Laplace. Across grids, with paired sequences, Student-t_5's local slope on the half-unit interval from $t = 4$ is -4.03 at $(801, 8)$, -5.92 at $(1601, 16)$ and -7.92 at $(3201, 32)$, against -8: the departure on the narrow domain is the truncated prior in the reference. Student-t_7 and t_9 behave the same way. The uniform family departs identically on every grid from $t \approx 3.5$: that is the double-precision floor of the posterior means, reached by every family near $t = 5$.

Moment estimators (e19) and selection without the true score (e20). On e06's data the moment estimator of ρ has RMSE 0.031, 0.039, 0.060 and 0.385 at $t = 0.05$, 0.4, 1 and 2; that of κ_4 has RMSE 1.42, 1.84 and 15.22 at $t = 0.05$, 0.4 and 1, with a bias of -0.75 at $t = 0.4$. Selecting EM–BP's checkpoint by held-out one-view log-evidence instead of by validation risk against the true kernel gives a test error of 0.105 against 0.105 at $M = 32$, the size at which selection matters most (7 of 8 seeds select the same rung; at $M = 2048$ the validation curve is monotone through the ladder, so every rule picks its top).

The information curve at $\rho = 0.5$ and the scalar constants (e21). At $\rho = 0.5$ the local slope of $\log(1/I^{\mathrm{eff}})$ on the levels $[0.6, 1.0]$ is 6.1–6.9 for the shape coordinate and 3.8–3.8 for the correlation, against the asymptotic 8 and 4: a sample-complexity experiment on those levels must be read against the curve, not the asymptote. The Cramér–Rao reference for the excess kurtosis at $t = 1$ is 1.62 at $M = 2048$ and 0.81 at $M = 8192$. In the scalar generalised-Gaussian family an untruncated quadrature gives $I_t(\beta)e^{8t}$ over $\frac{1}{24}(\partial_\beta \kappa_4)^2$ equal to 1.0000 ($\beta = 2$), 1.0009 ($\beta = 3$) and 0.9994 ($\beta = 1.5$) at $t = 2$, and the variance coordinate's $I_t e^{4t}$ equal to 1.0000 times one half.